

1.08 Radiation-Induced Segregation[☆]

Maylise Nastar and Frédéric Soisson, The French Alternative Energies and Atomic Energy Commission, Gif-sur-Yvette, France

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Abbreviations

1nn	first nearest neighbor	MIK	modified inverse Kirkendall
2nn	second nearest neighbor	nn	nearest neighbor
AKMC	atomic kinetic Monte Carlo	NRT	Norgett, Robinson, and Torrens
bcc	body-centered cubic	PPM	path probability method
DFT	density functional theory	RIP	radiation-induced precipitation
dpa	displacement per atom	RIS	radiation-induced segregation
fcc	face-centered cubic	SCMF	self-consistent mean field
IASCC	irradiation-assisted stress corrosion cracking	TEM	transmission electron microscopy
IK	inverse Kirkendall	TIP	thermodynamics of irreversible processes

Symbols

D	Diffusion coefficient	L_{ij}, L or L-coefficient	Phenomenological coefficient
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1.08.1 Introduction

Irradiation creates excess point defects in materials (vacancies and self-interstitial atoms), which can be eliminated by mutual recombination, clustering, or annihilation on preexisting or radiation-induced defects in the microstructure, such as surfaces, grain boundaries, dislocations, or point defect clusters (dislocation loops and voids). As a result, permanent irradiation sustains fluxes of point defects toward these point defect sinks and, in case of any preferential transport of one of the alloy components, leads to a local chemical redistribution. These radiation-induced segregation (RIS) phenomena are very common in alloys under irradiation and have important technological implications. Specifically in the case of austenitic steels, because Cr depletion at the grain boundary is suspected to be responsible for irradiation-assisted stress corrosion, a large number of experiments have been conducted on the RIS dependence on alloy composition, impurity additions, irradiation flux and time, irradiation particles (electrons, ions, or neutrons), annealing treatment before irradiation, and the nature of grain boundaries.^{1–5} Experimental observations of nuclear materials at low and high radiation doses tend to prove that RIS on point defect clusters are precursors to radiation-induced precipitation of secondary phases.⁶ RIS modeling is therefore a necessary step in order to obtain a good understanding of the materials stability under irradiation.

The first RIS models consisted in adapting the Fick's laws to reproduce two specific effects of irradiation: the enhancement of diffusion due to the increase of point defect concentration, and the emergence of new driving forces associated with the point defect concentration gradients. According to these models, the kinetic coefficients D or L (defined below) relating atomic fluxes to gradients of concentration or chemical potential determine RIS. The framework of the thermodynamics of irreversible processes (TIPs) within the linear response theory yields the best definition of these kinetic coefficients. Because the methods of calculating kinetic coefficients and the associated difficulties are different in a dilute or concentrated alloy, since the beginning, there have been two types of RIS models, those dedicated to dilute alloys and those to concentrated alloys.

Very recently, the dilute alloy models have benefited from progress made in the diffusion theories and from high throughput DFT calculations of the point defect jump frequencies.⁷ There are now automatic codes providing the phenomenological coefficients L in function of the atomic jump frequencies in multi-component dilute alloys, including the split-interstitial (dumbbell) diffusion mechanism.⁸ These codes allow the immediate use of extensive *ab initio* atomic jump frequency databases for a quantitative and systematic estimation of the flux couplings and sign of RIS with respect to temperature.^{9–12}

While the progress of RIS models of dilute alloys is closely related to that of diffusion theory, most segregation models for concentrated alloys still use oversimplified diffusion models based on Manning's relations.¹³ There are several reasons for this. First, the parameterization of an atomic jump frequency model in function of the surrounding alloy composition requires a large database of experimental diffusion coefficients, which are lacking and/or numerous *ab initio* calculations of atom jump frequencies. Finally, the jump sequences of the atoms are particularly complex in a multicomponent alloy, on account of the multiple jump frequencies and correlation effects that are involved. Only recently has an interstitial diffusion model been developed that

could account for short-range order effects, including binding energies with point defects in Fe–Cr concentrated alloys.¹⁴ Numerous studies devoted to RIS in concentrated systems are experimental works, investigating the effect of temperature, composition of model and technological alloys, irradiation conditions, as well as the effect of point defect trapping by an impurity, and the nature and displacement of a specific grain boundary. Most of the diffusion models of concentrated alloys rely on tracer diffusion data, from which one estimates the partial diffusion coefficients of the RIS diffusion equation. Paradoxically, the first RIS models were more rigorous¹⁵ than the present ones in which thermodynamic activities, particularly some of the cross-terms, are oversimplified. In this review, we go back to the first models starting from the linear response theory, albeit slightly modified, to be able to reproduce the main characteristics of an irradiated alloy.

Atomistic simulation methods such as atomic kinetic Monte Carlo (AKMC) methods^{14,16–20} have become efficient tools to simulate RIS. Thanks to a better knowledge of jump frequencies after the recent developments of DFT calculations, these simulations provide a fine description of the thermodynamics as well as the kinetics of a specific alloy. Moreover, information at the atomic scale is precious when RIS profiles exhibit oscillating behavior and spread over a few tens of nanometers. At the upper scale, Phase Field methods start addressing the complex issue of the impact of the microstructure – point defect interaction on RIS, including long-range elastic interactions between defects.

In the second section, we present discoveries and typical observations of RIS. In the third section, we rely on the formalism of TIP to write the alloy flux couplings. In section four, after a short review on available thermodynamic and diffusion data, we discuss the approximations and limitations of empirical and atomistic RIS models applied to the analysis of real systems. We then present the alternative approach based on atomistic modeling of jump mechanisms. The last section introduces multi-scale simulation techniques of RIS.

1.08.2 Experimental Observations

1.08.2.1 Anthony's Experiments

RIS was predicted by Anthony²¹ in 1969, a few years before the first experimental observations: a rare case in the field of radiation effects. The prediction stemmed from an analogy with nonequilibrium segregation observed in aluminum alloys quenched from high temperature. Between 1968 and 1970, in a pioneering work in binary aluminum alloys, Anthony and coworkers^{21–25} systematically studied the nonequilibrium segregation of various solute elements on the pyramidal cavities formed in aluminum after quenching from high temperature. They explained this segregation by a coupling between the flux of excess vacancies toward the cavities and the flux of solute (Fig. 1). Nonequilibrium segregation had been previously observed by Kuczynski *et al.*²⁶ during the sintering of copper-based particles and by Aust *et al.*²⁷ after the quenching of zone refined metals.

Anthony suggested that similar coupling should produce nonequilibrium segregation in alloys under irradiation.^{21,22} He predicted that the segregation should be much stronger than after quenching because under irradiation, the excess vacancy concentration and the resulting flux can be sustained for very long times.^{22,28} As for the cavities formed by vacancy condensation in alloys under irradiation, which result in the swelling phenomenon, he pointed out that with solute and solvent atoms of different sizes, segregation should generate strains around the voids.²⁸ Finally, he predicted intergranular corrosion in austenitic steels and zirconium alloys, resulting from possible solute depletion near grain boundaries.²⁸

Anthony also presented a detailed discussion on nonequilibrium segregation mechanisms, in the framework of the TIP,^{21–24} showing that the nonequilibrium tendencies are controlled by the phenomenological coefficients L_{ij} of the Onsager matrix, which can be – in principle – computed from vacancy jump frequencies (see below Section 1.08.3). Clarifying previous discussions on non-equilibrium segregation mechanisms,^{26,27} he considered two limiting cases for the coupling between solute

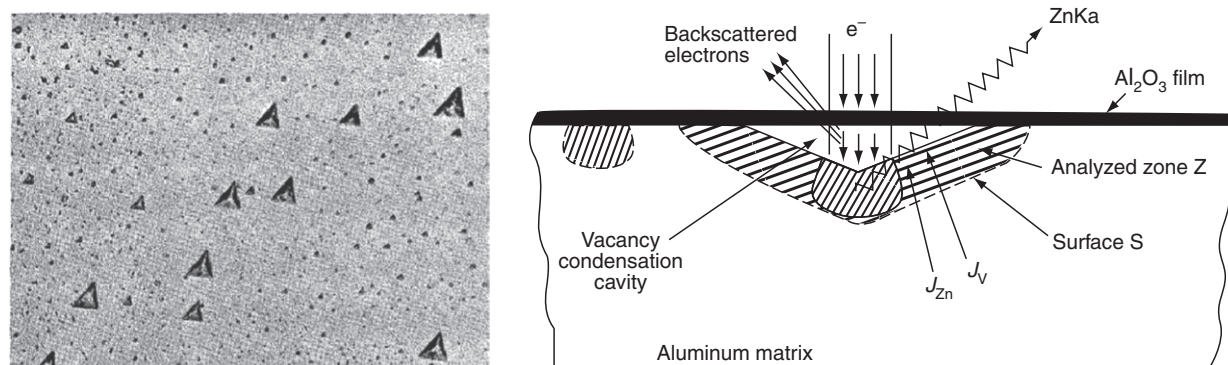


Fig. 1 One of Anthony's experiments. After quenching of an Al–Zn alloy, vacancies condense in small pyramidal cavities (left), under an Al_2O_3 thin film covering the surface. Electronic probe measurements reveal enrichments in Zn around the cavities. Reproduced from Anthony, T.R., 1970. J. Appl. Phys. 41, 3969–3976.

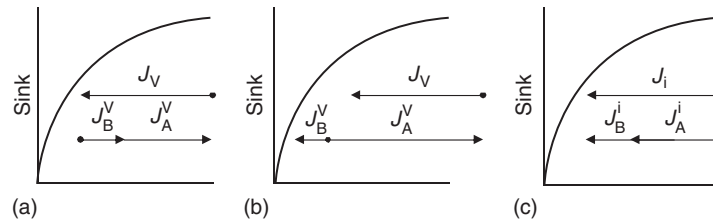


Fig. 2 Radiation-induced segregation mechanisms due to coupling between point defect and solute fluxes in a binary A–B alloy. (a) An enrichment of B occurs if $d_{BV} < d_{AV}$ and a depletion if $d_{BV} > d_{AV}$. (b) When the vacancies drag the solute, an enrichment of B occurs. (c) An enrichment of B occurs when $d_{Bi} > d_{Ai}$.

and vacancy fluxes in an A–B alloy (at the time, he did not apparently consider the coupling between solute and interstitial fluxes and its possible contribution to RIS). In both cases, the total flux of atoms must be equal and in the direction opposite to the vacancy flux:

- (1) If both A and B fluxes are in the direction opposite to the vacancy flux (**Fig. 2(a)**), one can expect a depletion of B near the vacancy sinks if the partial diffusion coefficient of B is larger than that of A ($d_{BV} > d_{AV}$); in the opposite case ($d_{BV} < d_{AV}$), one can expect an enrichment of B (this was essentially the explanation proposed by Kuczynski *et al.*²⁶). Note that as long as the solute concentration is not too small (C_B roughly below 1 at%), the tracer diffusion coefficients, $D_{A^*}^V$ and $D_{B^*}^V$, divided by the equilibrium vacancy concentration, provide a good estimation of the partial diffusion coefficients, d_{AV} and d_{BV} (cf Section 1.08.3.2. for a definition of d_{AV} and d_{BV}).
- (2) However, A and B fluxes are not necessarily in the same direction. Whenever, the B solute atoms are strongly bound to the vacancies, there is a chance vacancy drags a solute atom B by following circular jump sequences around B between two exchanges with B. Then, the vacancy and solute fluxes can be in the same direction (**Fig. 2(b)**): this was the explanation proposed by Aust *et al.*²⁷ In such a case, an enrichment of B is expected, even if the amplitude of d_{BV} is higher than d_{AV} . In that case, the partial diffusion coefficient d_{BV} is negative and cannot be related to $D_{B^*}^V$.

Starting from an initially uniform solid solution, these mechanisms lead to formation of a gradient in solute concentration ∇C_B at the sinks, when the ratio of the total fluxes J_B/J_A differs from C_B/C_A . In turn, ∇C_B produces a back diffusion flux, according to Fick's law. The competition between these two driving forces leads to steady state RIS concentration profiles. Unlike Anthony's quenching experiments, segregation under permanent irradiation is not a transient state. RIS profiles are sustained as long as there is no drastic change of the sink microstructure. Indeed, the segregation of elemental species driven by the point defect concentration gradients, yields concentration gradients of the elemental species at the region of point defect sinks. In turn, chemical driving forces arise to counter the built-up gradients resulting from the point defect concentration gradient.

1.08.2.2 First Observations of RIS

In 1972, Okamoto *et al.*²⁹ observed strain contrasts around voids in an austenitic stainless steel Fe–18Cr–8Ni–1Si during irradiation in a high-voltage electron microscope. They attributed this contrast to the segregation strains predicted by Anthony. This is the first reported experimental evidence of RIS. Soon after, a chemical segregation was directly measured by Auger spectroscopy at the surface of a similar alloy irradiated by Ni ions.³⁰

It was soon realized that if the solute concentration near the point defect sinks reaches the solubility limit, a local precipitation would take place. In 1975, Barbu and Ardell³¹ observed such a radiation-induced precipitation (RIP) of an ordered Ni₃Si phase in an undersaturated Ni–Si alloy.

The analysis of strain contrast and concentration profiles measured by Auger spectroscopy suggested that undersized Ni and Si atoms (which can be more easily accommodated in interstitial sites) were diffusing toward point defect sinks, while oversized atoms (such as Cr) were diffusing away. Such a trend, later confirmed in other austenitic steels and nickel-based alloys,³² led Okamoto and Wiedersich³⁰ to conclude that RIS in austenitic steels was due to the migration of interstitial–solute complexes, and they proposed this new RIS mechanism, in addition to the ones involving vacancies (**Fig. 2(c)**). Then again, Marwick³³ explained the same experimental observations by a coupling between fluxes of vacancies and solute atoms, pointing out that thermal diffusion data showed Ni to be a slow diffuser and Cr to be a rapid diffuser in austenitic steels. We will see later that, in spite of many experimental and theoretical studies, the debate on the diffusion mechanisms responsible for RIS in austenitic steels is not over.

Following these debates on RIS mechanisms, it became common to refer to the situation illustrated in **Fig. 2(a)** as segregation by an inverse Kirkendall (IK) effect (the term was coined by Marwick³³ in 1977) and to the one in **Fig. 2(b)** as segregation by solute drag effects, or by migration of vacancy–solute complexes. In the classical Kirkendall effect,³⁴ a gradient of chemical species produces a flux of defects. It occurs typically in interdiffusion experiments in A–B diffusion couples, when A and B do not diffuse at the same speed. A vacancy flux must compensate for the difference between the flux of A and B atoms, and this leads to a shift of the initial A/B interface (the Kirkendall plane). The IK effect is due to the same diffusion mechanisms but corresponds to the

situation where the gradient of point defects is imposed and generates a flux of solute. The distinction between RIS by IK effect and RIS by migration of defect–solute complexes, initially proposed for the vacancy mechanisms, was soon generalized to interstitial fluxes by Okamoto and Rehn.^{35,36} RIS in dilute alloys, where solute–defect binding energies are clearly defined and often play a key role, is commonly explained by diffusion of solute–defect complexes, while the IK effect is often used to explain RIS in concentrated alloys. This distinction is reflected in the modeling of RIS (see Section 1.08.3). However, it is clear that RIS can occur in dilute alloys without migration of solute–defect complexes. Moreover, such a terminology and sharp distinction can be somewhat misleading; the mechanisms are not mutually exclusive and result from the same microscopic diffusion mechanisms. In the case of undersized B atoms, for example, a strong binding between interstitial and B atoms can lead to a rapid diffusion of B by the interstitial (IK effect with $d_{BI} > d_{AI}$) and to the migration of interstitial–solute complexes. More generally, one can always say that RIS results from an IK effect, in the sense that it occurs when a gradient of point defects produces a flux of solute. Nevertheless, because they are widely used, we will refer to these terms at times when they do not create confusion.

1.08.2.3 General Trends

Many experimental studies of RIS were carried out in the 1970s in model binary or ternary alloys, as well as in more complex and technological alloys (especially in stainless steels). It became apparent quite early on that RIS was a pervasive phenomenon, occurring in many alloys and with any kind of irradiating particle (ions, neutrons, or electrons). Extensive reviews can be found in Russell,¹ Holland *et al.*,² Nolfi,³ Ardell,⁴ Was,⁵ Ardell and Bellon³⁷; here, we present only the general conclusions that can be drawn from these studies.

1.08.2.3.1 Segregating elements

From the previous discussion, it is clear that it is difficult to predict the segregating element in a given alloy because of the competition between several mechanisms and the lack of precise diffusion data (especially concerning interstitial defects). As will be shown in Section 1.08.3, only the knowledge of the phenomenological coefficients L_{ij} provides a reliable prediction of RIS. Nevertheless, relying on the body of RIS experimental studies, several general rules have been proposed. In dilute binary AB alloys, thermal self-diffusion coefficients D_A^V and impurity diffusion coefficients D_B^V are generally well known, at least at high temperatures. Tracer diffusion or intrinsic diffusion coefficients in some concentrated alloys are also available.³⁸ RIS experiments do not reveal a systematic depletion of the fast-diffusing and enrichment of the slow-diffusing elements near the point defect sinks;^{4,32} this suggests that the IK effect by vacancy diffusion is usually not the dominant mechanism. On the other hand, a clear correlation exists between RIS and the size effect³⁶; undersized atoms usually segregate at point defect sinks, oversized atoms usually do not. This suggests that interstitial diffusion could control the RIS, at least for atoms with a significant size effect. There are some exceptions: in Ni–Ge and Al–Ge alloys, the segregation of oversized solute atoms has been observed. Nevertheless, as pointed out by Rehn and Okamoto,³⁶ no case of depletion of undersized solute atoms in dilute alloys has ever been reported. According to Ardell,⁴ this holds true 25 years later.

1.08.2.3.2 Segregation profiles

Experiments show that segregation concentration profiles induced by irradiation display some specific features. They can spread over large distances – a few tens of nanometers (see Figs. 7 and 8 and other examples in Russell¹ and Okamoto and Rehn³²) – while equilibrium segregation is usually limited to a few angstroms. This is because they result from a dynamic equilibrium between RIS fluxes and the back diffusion created by the concentration gradient at the sinks, while the scale of equilibrium segregation profiles is determined by the range of atomic interactions. Equilibrium profiles are usually monotonic, except for the oscillations, which can appear – with atomic wavelengths – in alloys with ordering tendencies.³⁹ Segregation profiles observed in transient regimes are often non-monotonic because of the complex interaction between concentration gradients of point defects and solutes. A typical example is shown in Section 1.08.5.3, where an enrichment of solute is observed near a point defect sink, followed by a smaller solute depletion between the vicinity of sink and the bulk. In this particular case, the depletion is due to a local increase in vacancy concentration, which results from a lower interstitial concentration and recombination rate.

Other kinds of non-monotonic profiles are sometimes observed, with typical ‘W-shapes’.⁴⁰ In some austenitic or ferritic steels, a local enrichment of Cr at grain boundaries survives during the Cr depletion induced by irradiation (see below). This could result from a competition between opposite equilibrium and RIS tendencies. However, the extent of the Cr enrichment often seems too wide to be simply due to an equilibrium property (around 5 nm, see, e.g., Sections 1.08.2.5 and 1.08.5.3).

RIS profiles at grain boundaries are sometimes asymmetrical, which has been related to the migration of boundaries resulting from the fluxes of point defects under irradiation.^{41,42} The segregation is affected by the atomic structure and the nature of the sinks. It has been clearly shown that RIS in austenitic steels is smaller at low angles and special grain boundaries than at large misorientation angles,^{43,44} the latter being more efficient point defect sinks than the former.

1.08.2.3.3 Temperature effects

RIS can occur only when significant fluxes of defects towards sinks are sustained, which typically happens only at temperatures between 0.3 and 0.6 times the melting point. At lower temperatures, vacancies are immobile and point defects annihilate mainly by mutual recombination. At higher temperatures, the equilibrium vacancy concentration is too high; back diffusion and a lower

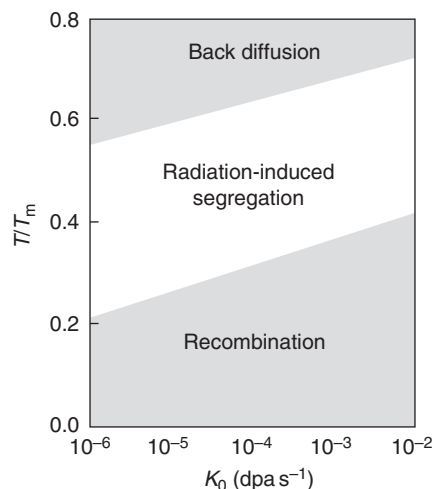


Fig. 3 Temperature and dose rate effect on the radiation-induced segregation.

vacancy supersaturation completely suppress the segregation. Temperature can also modify the direction of the RIS by changing the relative weight of V and SIA competing mechanisms, which do not have the same activation energy. We expect this competition occurs in Fe–Cr alloys (cf Section 1.08.4.4.3). In Ni–Ti alloys, the enrichment of Ti at the surface below 400°C has been ascribed to the migration of Ti–V complexes, and the depletion observed at higher temperature should result from a vacancy IK effect.⁴⁵

1.08.2.3.4 Effects of radiation particles, dose, and dose rates

RIS can be observed for very small irradiation doses; an enrichment of ~10% of Si has been measured, for example, at the surface of an Ni–1%Si alloy, after a dose of 0.05 dpa at 525°C.³⁵ Such doses are much lower than those required for radiation swelling⁵ or ballistic disordering effects.⁴⁶

Increasing the radiation flux, or dose rate (Φ), directly results in higher point defect concentrations and fluxes towards sinks. The transition between RIS regimes is then shifted toward a higher temperature (Fig. 3). But because point defect concentrations slowly evolve with the radiation flux (typically proportional to Φ or $\sqrt{\Phi}$ ⁴⁷ in the temperature range where RIS occurs), a high increase is needed to get a significant temperature shift. However, within a given kinetic regime, the behavior of RIS differs from that of a point defect concentration. For example, one has shown that in the kinetic regime dominated by point defect elimination at sinks, the amount of RIS does not dependent on the radiation flux as long as the point-defect sink strength does not vary.²⁰ Radiation dose and dose rate are usually estimated in dpa and dpa s⁻¹, respectively, using the Norgett, Robinson, and Torrens model,⁴⁸ especially when a comparison between different irradiation conditions is desired. It is then worth noting that the amount of RIS observed for a given dpa is usually larger during irradiation by light particles (electrons or light ions) than by heavy ones (neutrons or heavy ions). In the latter case, point defects are created by displacement cascades in a highly localized area, and a large fraction of vacancies and interstitials recombine or form point defect clusters. The fraction of the initially produced point defects that migrate over long distances and could contribute to RIS is reduced. On the contrary, upon irradiation by light particles, the creation of Frenkel pairs is more or less homogeneous in the material, and most of them survive and migrate as isolated point defects.⁴⁹

1.08.2.3.5 Impurity effects

The addition of impurities has been explored as a possible way to control RIS in alloys, for example, in austenitic steels. The most common method is the addition of an oversized impurity, such as Hf and Zr, in stainless steels,^{50,51} which should trap the vacancies (and, in some cases, the interstitials), thus increasing the mutual point defect recombination reactions at the expense of elimination reactions at sinks. However, the experimental investigations tend to show the suppressing of RIS by an impurity lasts for a few displacements per atom only. This is certainly due to the kinetic drag of impurities by point defects toward sinks. A recent DFT study demonstrates it almost systematically occurs whenever there is a large positive point defect-impurity binding energy, at least in dilute FCC solid solutions.¹¹

1.08.2.3.6 Density of sinks

In the search for radiation tolerant materials, one devises microstructures with a high density of sinks as for example bulk nano-layered Cu–Nb composites containing interfaces with controlled sink efficiencies.⁵² Multiplying point defect sinks reduce the amount of RIS at a given sink. A simplified RIS analytical model of multilayers separated by sink interfaces demonstrates the amount of segregation decreases linearly with decreasing interface spacing.²⁰

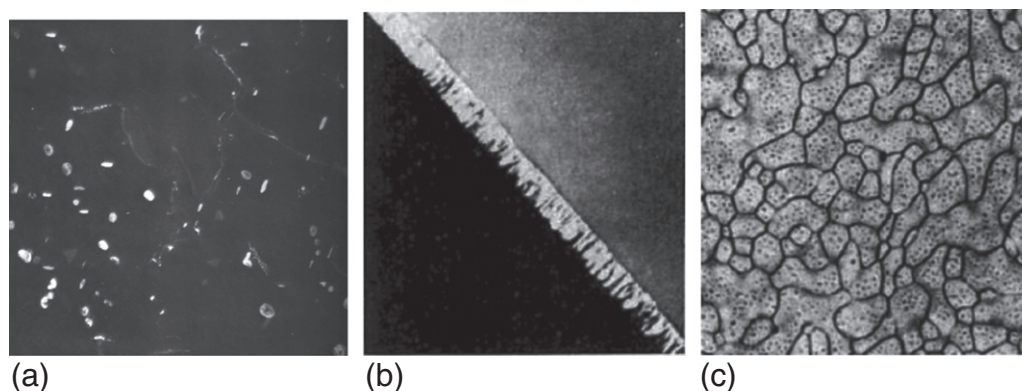


Fig. 4 Formation of Ni_3Si precipitates in undersaturated solid solution under irradiation (a) in the bulk on preexisting dislocations and at interstitial dislocations, (b) at grain boundaries, and (c) at free surfaces. (a) Courtesy of A. Barbu. Reproduced from Holland, J.R., Mansur, L.K., Potter, D.I., 1981. *Phase Stability During Radiation*. Warrendale, PA: TMS-AIME.

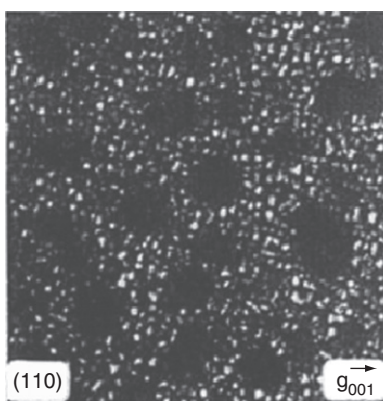


Fig. 5 Dissolution of γ' precipitates near dislocation loops in Ni-Al under irradiation. Reproduced from Holland, J.R., Mansur, L.K., Potter, D.I., 1981. *Phase Stability During Radiation*. Warrendale, PA: TMS-AIME.

1.08.2.4 RIS and Precipitation

As mentioned above, one of the most spectacular consequences of RIS is that it can completely modify the stability of precipitates and the precipitate microstructure.⁵³ When the local solute concentration at a point defect sink reaches the solubility limit, RIP can occur in an overall undersaturated alloy. RIP of the γ' - Ni_3Si phase is observed, for example, in Ni-Si alloys³¹ at concentrations well below the solubility limit (Ni_3Si is an ordered L_{12} structure and can be easily observed in dark-field image in transmission electron microscopy (TEM)). In this case, we believe RIS is due to the preferential occupation of interstitials by undersized Si atoms.³¹ The γ' -phase can be observed on the preexisting dislocation network, at dislocation loops formed by self-interstitial clustering,³¹ at free surfaces⁴⁹ or grain boundaries (Fig. 4(a)).⁵⁴ The fact that the γ' -phase dissolves when irradiation is stopped clearly reveals the nonequilibrium nature of the precipitation. This is also shown by the toroidal contrast of dislocation loops (cf Fig. 4): the γ' -phase is observed only at the border of the loop on the dislocation line where self-interstitials are annihilated; when the loop grows, the ordered phase dissolves at the center of the loop, which is a perfect crystalline region where no flux of Si sustains the segregation.

In supersaturated alloys, irradiation may completely modify the existing precipitation microstructure. It may dissolve precipitates located near sinks when RIS produces a solute depletion. For example, in Ni-Al alloys,⁵⁵ dissolution of γ' precipitates is observed around the growing dislocation loops, due to the Al depletion induced by irradiation (Fig. 5), and in supersaturated Ni-Si alloys, Si segregation towards the interstitial biased sinks produces dissolution of the homogeneous precipitate microstructure in the bulk, to the benefit of the precipitate layers on the surfaces³¹ (Fig. 6) and grain boundaries.⁵⁶

In these examples, RIS was observed to produce a heterogeneous precipitation at point defect sinks. But homogeneous RIP of coherent precipitates has also been observed, for example, in Al-Zn alloys.⁵⁷ Cauvin and Martin⁵⁸ have proposed a mechanism that explains such a decomposition. A solid solution contains fluctuations of composition. In case of attractive vacancy-solute and interstitial-solute interactions, a solute-enriched fluctuation tends to trap both vacancies and interstitials, thereby favoring mutual

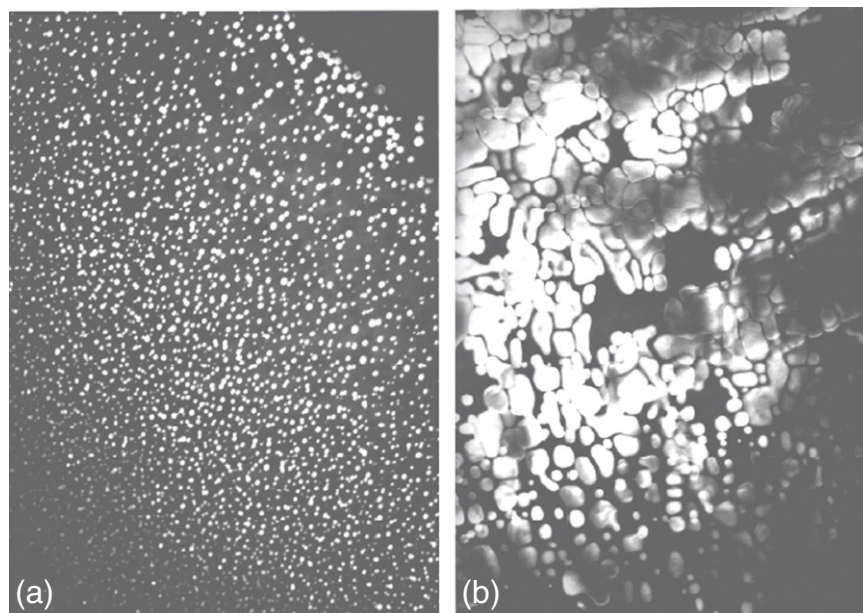


Fig. 6 Precipitate microstructure in Ni-Si alloys: the homogeneous distribution observed during thermal aging (a) is dissolved under electron irradiation and the surfaces of the transmission electron microscopy sample are covered by Ni_3Si precipitates. (b) Ni-12%Si alloy under 1 MeV electron irradiation at 500°C, after a dose of 5×10^{-5} dpa. Courtesy of A. Barbu.

recombination. The point defect concentrations then decrease, producing a flux of new defects toward the fluctuation. If the coupling with solute flux is positive, additional solute atoms arrive on the enriched fluctuations, and so it continues, until the solubility limit is reached.

1.08.2.5 RIS in Austenitic and Ferritic Steels

We have seen that RIS was first observed in austenitic steels on the voids that are formed at large radiation doses. The depletion of Cr at grain boundaries is suspected to play a role in irradiation-assisted stress corrosion cracking (IASCC); this is one of the many technological concerns related to RIS. The enrichment of Ni and the depletion of Cr can also stabilize the austenite near the sinks, and favor the austenite $\rightarrow$ ferrite transition in the matrix.³² The segregation of minor elements can lead to the heterogeneous precipitation of γ' -precipitates (as in Ni-Si alloys), or various M_{23}C_6 carbides and other phases.^{1,32}

The segregation of major elements always involves an enrichment of Ni and a depletion of Cr at sinks over a length scale that depends on the alloy composition and irradiation conditions (Fig. 7).⁵ Up to now, there is no definite answer on the RIS mechanism controlling the segregation of the major elements.⁵⁹ It is not clear whether it is driven by vacancy fluxes, as suggested by the order of the thermal tracer diffusion coefficients, $D_{\text{Ni}}^V < D_{\text{Fe}}^V < D_{\text{Cr}}^V$,³³ or by the migration of interstitial-solute complexes, resulting in the segregation of undersized atoms.³² Some models of RIS take into account only the first mechanism,⁵ while others predict a significant contribution of interstitials to the diffusion of Ni.^{60,61} For the segregation of minor elements, the size effect seems dominant, with an enrichment of undersized atoms (e.g., Si³⁰) and a depletion of oversized atoms (e.g., Mo⁶²).

The effect of minor elements on the segregation behavior of major ones has been pointed out since the first experimental studies³²; Si and Mo additions, by trapping vacancy, would enhance the vacancy-interstitial recombination rate which in turn would reduce the RIS of both major and minor elements. As previously mentioned, oversized impurity atoms, such as Hf and Zr, may also decrease RIS.⁵⁰

RIS in ferritic steels and ferritic/martensitic steels has recently drawn much attention, because they are possible candidates for future Generation IV and fusion reactors.⁶³ Experimental studies are more difficult than in austenitic steels, partly because of the complex pre-existing microstructures. The behavior of some minor elements has been clearly identified (such as the enrichment of undersized Si and P atoms^{63,64}). However, the segregation of major elements (Fe and Cr) is still debated. Early experimental studies reviewed by Lu *et al.*⁶⁵ have shown both depletion and enrichment of Cr at grain boundaries, but it appeared quite difficult to distinguish correlations between the RIS behavior and the irradiation conditions or the alloy compositions. In some highly concentrated alloys a depletion of Cr (sometimes with W-shaped profiles, see Fig. 8) and an enrichment of Ni have been observed,⁶³ reminding us of the general trends in austenitic steels. The addition of Hf has been found to reduce the Cr segregation, again similarly to what occurs in austenitic steels.⁶⁴

More recently, systematic studies by Was *et al.*^{59,66} focusing on Fe-Cr model and industrial alloys with Cr contents between 9 and 12 at% generally displayed Cr enrichment at intermediate temperatures, and Cr depletion at higher temperature. The increase in Cr content would favor the depletion. These behaviors are consistent with the predictions of recent modeling studies. They

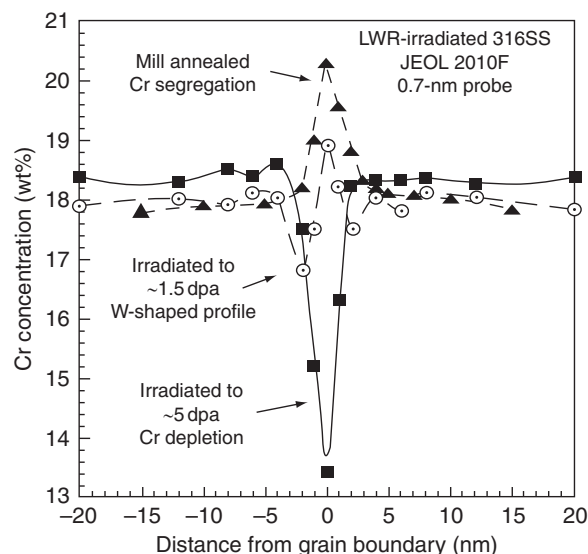


Fig. 7 Thermal and radiation-induced segregation profiles in 316 stainless steel. Reproduced from Bruemmer, S.M., Simonen, E.P., Scott, P.M., Andresen, P.L., Was, G.S., Nelson, J.L., 1999. *J. Nucl. Mater.* 274, 299–314.

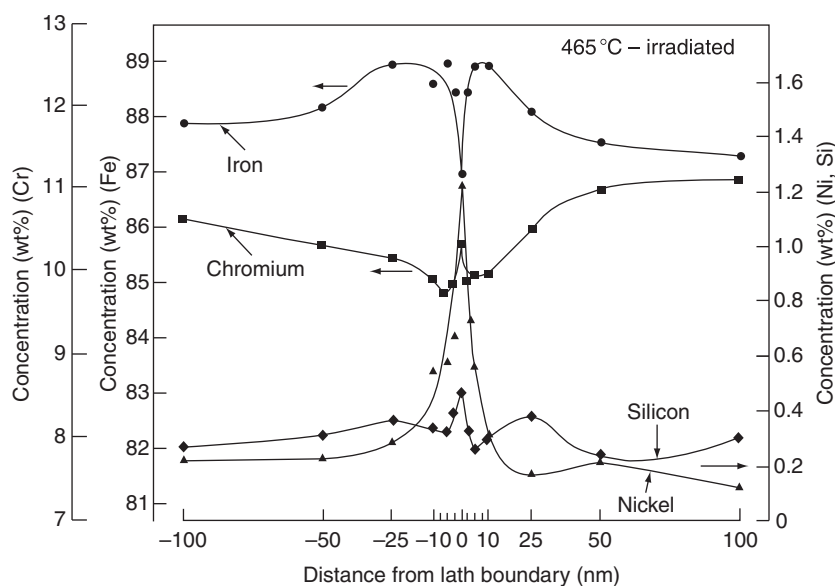


Fig. 8 Concentration profiles of Cr, Ni, Si, and Fe on either side of a lath boundary in 12%Cr martensitic steel after neutron irradiation to 46 dpa at 465°C. Reproduced from Little, E., 2006. *Mater. Sci. Technol.* 22, 491–518.

would result from the competition between vacancies and self-interstitials (cf Section 1.08.5.3). However, many observations do not fit this simple scenario, possibly because other factors may be involved. For example, it has been observed that the enrichment of Cr at grain boundaries could increase up to a few dpa and decrease at larger doses⁶⁶ (a possible effect of the long term evolution of the sink microstructure). Minor elements (especially C) may co-segregate with Cr,⁶⁷ or lead to the precipitation of other phases affecting the Cr segregation.⁶⁸ The orientation and structure of grain boundaries^{67,69} have also an effect. All these factors have a stronger impact than in austenitic steels, leading to less predictable behaviors.

1.08.2.6 RIS in High Entropy Alloys

High entropy alloys, with equal or relatively large proportions of five or more elements have recently received considerable attention, and a few experimental studies of RIS have been published.^{70–74}

Segregation at radiation-induced dislocation loops in CrFeCoNi, CrFeCoNiMn, and CrFeCoNiPd alloys have been observed during electron irradiation, showing depletions in Cr/Fe/Mn/Pd and enrichments in Co/Ni.⁷⁰ The authors relate this behavior to the atomic size of the elemental species. The formation of a L1₀ (NiMn) ordered phase and a spinodal decomposition between Co/Ni and Pd (also observed during long term annealing) is accelerated by irradiation.

Barr *et al.* have observed Mn depletion and Co and Ni enrichment at grain boundaries, upon ion irradiation of an equiatomic CoCrFeNiMn.⁷¹ According to tracer diffusion experiments, Mn is the fastest vacancy diffuser in these alloys and Co is the slowest diffuser. These observations are consistent with a RIS mechanism dominated by a vacancy flux coupling, the latter being modeled through an inverse Kirkendall process. Such a RIS mechanism is reminiscent of what is suggested in most of the RIS models of austenitic steels.

1.08.3 Diffusion Equations: The TIP Formalism

RIS is a phenomenon that couples the fluxes of defects created by irradiation to the atomic fluxes of an alloy. In standard models, only point defect diffusion mechanisms are considered, although displacement cascades produce mobile point defect clusters, which may contribute to RIS. A RIS continuous model relies on diffusion-reaction equations modeling the evolution of vacancy, interstitial and the elemental species of the alloy.

Far from sinks, we may transform a diffusion-and-annihilation at sinks process into a simple reaction event. For instance, in a standard mean field homogeneous rate theory, the evolutions of vacancy and self-interstitial atomic fractions (Confusions in the definition of concentrations are a source of frequent mistakes. In the following, unless explicitly specified, the concentrations of a specie α is given in atomic (or molar) fraction, C_α . The atomic concentration, i.e., the number of atoms per unit volume, is C_α/Ω , where Ω is the atomic volume), C_V and C_I are given by

$$\begin{aligned}\frac{dC_V}{dt} &= K_0 - RC_I C_V - \sum_s k_{Vs}^2 D_V (C_V - C_V^{eq}) \\ \frac{dC_I}{dt} &= K_0 - RC_I C_V - \sum_s k_{Is}^2 D_I C_I\end{aligned}\quad (1)$$

where K_0 is the point defect production rate (in dpa s⁻¹) proportional to the radiation flux, R is the recombination rate, and D_V and D_I are the point defect diffusion coefficients. The third terms of the right hand side in Eq. (1) correspond to point defect diffusion and annihilation at sinks of type s . The 'sink strengths' k_{Vs}^2 and k_{Is}^2 depend on the nature and the density of sinks and have been calculated for all common sinks, such as dislocations, cavities, free surfaces, grain boundaries, etc.^{75,76} Numerical integration of Eq. (1) yields the evolution of point defect concentrations and their variation with the radiation flux and sink microstructure. We may as well derive steady state analytical solution of Eq. (1).⁴⁷

A homogenous mean field theory is not sufficient to model flux couplings and subsequent RIS. We must start from the writing of the diffusion-annihilation at sinks in terms of divergence of fluxes J . The evolution of atomic fractions of vacancies, interstitials, and chemical elements α in an alloy under irradiation is given by

$$\begin{aligned}\frac{\partial C_V}{\partial t} &= -\text{div } J_V + K_0 - RC_I C_V - \sum_s k_{Vs}^2 D_V [C_V - C_V^{eq}] \\ \frac{\partial C_I}{\partial t} &= -\text{div } J_I + K_0 - RC_I C_V - \sum_s k_{Is}^2 D_I C_I \\ \frac{\partial C_\alpha}{\partial t} &= -\text{div } J_\alpha\end{aligned}\quad (2)$$

The basic problem of RIS is to find the solution of these equations close to a given point defect sink, which requires the knowledge of how the fluxes J_α are related to the concentrations. Note that the last terms of the point defect rate equations represent diffusion-annihilation reactions of point defects at the other sinks, which are treated within a mean field homogeneous approximation.

The modeling of the atomic and point defect transport controlling fluxes relies on the theory of TIP. In this article, we start with a general description of TIP applied to transport. We write atomic fluxes in terms of the phenomenological coefficients of diffusion (denoted hereafter by L_{ij} or, simply, L) and the driving forces $\nabla\mu$. Since what we measure and follow are the on-site average concentrations of the elemental species, we express gradients of chemical potential, $\nabla\mu$, as function of gradients of concentration, ∇C . The terms of proportionality between fluxes and gradients of concentration are the so-called diffusion coefficients. Then, we deduce steady state relationship between gradients of concentration of the elemental species and the vacancy gradient of concentration.

1.08.3.1 Fluxes in Terms of L and $\nabla\mu$

Within the TIP,^{77,78} a system is divided into cells, which are supposed to be small enough to be considered as homogeneous and large enough to be in local equilibrium. The number of particles in a cell varies if there is a transfer of particles to other cells. The

transfer of particles α between two cells is described by a flux J_α , and the temporal variation of the local atomic fraction of α is given by the continuity equation

$$\frac{\partial C_\alpha}{\partial t} = -\operatorname{div} J_\alpha \quad (3)$$

The flux of species α between cells i and j is assumed to be a linear combination of the thermodynamic forces, $X_\beta = (\mu_\beta^j - \mu_\beta^i)/k_B T$, where μ_β^i is the chemical potential of species β in cell i , T the temperature, and k_B the Boltzmann constant. In the continuous limit, X_β is obtained from the gradient of chemical potential: $X_\beta = \nabla \mu_\beta / k_B T$. Variables X_β represent the deviation of the system from equilibrium, whose fluxes tend to reduce:

$$J_\alpha = - \sum_\beta L_{\alpha\beta} X_\beta \quad (4)$$

The equilibrium constants are the phenomenological (or Onsager) L -coefficients. Since the point defect diffusion mechanisms satisfy the detailed balance, the matrix of the Onsager coefficients is symmetric. Due to the second principle of thermodynamics, the production of entropy is positive and consequently the Onsager matrix ($L_{\alpha\beta}$) is definite positive. By construction, the vacancy diffusion mechanism leads to the relation between fluxes:

$$J_V = - \sum_\alpha J_\alpha^V. \quad (5)$$

Eq. (5) yields relationships between the phenomenological coefficients. For instance, if we choose to eliminate the $L_{V\beta}^V$ coefficients, we obtain this expression of the atomic fluxes:

$$J_\alpha^V = - \sum_\beta L_{\alpha\beta}^V (X_\beta - X_V). \quad (6)$$

Under irradiation, diffusion is controlled by both vacancies and interstitials. By construction, the interstitial diffusion mechanism leads to the relation between fluxes:

$$J_I = \sum_\beta J_\beta^I. \quad (7)$$

One assumes, vacancy and interstitial contributions to the atomic fluxes are additive:

$$J_\alpha = - \sum_\beta L_{\alpha\beta}^V (X_\beta - X_V) - \sum_\beta L_{\alpha\beta}^I (X_\beta + X_I). \quad (8)$$

The total flux of species α is thus expressed in terms of two Onsager matrices of size $N_{sp} \times N_{sp}$ where N_{sp} is the number of atomic species. Since every Onsager matrix is symmetric, it has $N_{sp} \times (N_{sp} - 1)/2$ independent L -coefficients. While thermodynamic data are usually available for the determination of driving forces in terms of gradients of concentration, it is very difficult to determine the whole set of L -coefficients from diffusion data. Experimentally, we do not measure couples of (L -coefficient, $\nabla \mu$) but couples of (D -coefficient, ∇C). In the next section, we define the driving forces as a function of concentration gradients, ∇C . Then, we define the experimental diffusion coefficients, D , in terms of the L -coefficients and thermodynamic driving forces. The last part of the section shows how to use first-principle calculations for building atomic jump frequency models to calculate macroscopic fluxes of specific alloys.

1.08.3.2 Fluxes in Terms of D and ∇C

Gradients of chemical potentials are the thermodynamic driving forces, which inform on the intensity of gradients of concentrations but also on the energy interactions between species. The latter make a chemical potential of a given species depend on concentrations of the other species. In the first part of the present article, we rely on the Gibbs–Duhem relationship and a conservation law to express the $\nabla \mu$ – ∇C relationship. Then, we deduce the steady state RIS relationship between gradients of concentration.

1.08.3.2.1 $\nabla \mu$ – ∇C relationship

The chemical potential of species i corresponds to the partial derivative of the Gibbs free energy G of the alloy, with respect to the number of atoms of species i , that is, N_i . The resulting chemical potential is a function of temperature and atomic fractions of the alloy components, $C_i = N_i/N$, N being the total number of atoms.

To obtain the $\nabla \mu$ – ∇C relationship, we rely on the Gibbs–Duhem relationship⁷⁷

$$\sum_{k=1,r} C_k \nabla \mu_k = 0, \quad (9)$$

and the conservation law

$$\sum_{k=1,r} C_k = 1, \quad (10)$$

where the sum runs over the number of species including point defects, though in front of the interstitial concentration we need to

add a minus sign. To deduce relationships between gradients of chemical potentials, we introduce either vacancies or interstitials into the alloy. Indeed, we ignore the interactions between I and V and assume their interactions with atomic species are additive. These interactions should be negligible because whenever I and V are close to each other, they annihilate through mutual recombination reaction. From Eqs. (9) and (10), we deduce chemical potential gradient of atomic species i in terms of the alloy and point defect driving forces

$$\begin{aligned}\nabla(\mu_i - \mu_V) &= - \sum_{k=1, k \neq i}^{N_p-1} \frac{C_k}{1 - C_V} \nabla(\mu_k - \mu_i) - \frac{1}{1 - C_V} \nabla\mu_V \\ \nabla(\mu_i + \mu_I) &= - \sum_{k=1, k \neq i}^{N_p-1} \frac{C_k}{1 + C_I} \nabla(\mu_k - \mu_i) - \frac{1}{1 + C_I} \nabla\mu_I\end{aligned}\quad (11)$$

Note that in thermal conditions, we usually ignore gradients of point defect chemical potential by assuming point defects are at local equilibrium with respect to the atomic species. Whereas, upon irradiation these gradients become the main driving forces.

From the Gibbs free energy, we deduce the alloy and point defect driving forces in function of concentration gradients. To compute the alloy chemical potential gradients $\nabla(\mu_k - \mu_i)$ of species k relative to species i , we ignore the point defect contributions. The latter is then equal to

$$\frac{\nabla(\mu_k - \mu_i)}{k_B T} = \frac{1}{k_B T} \sum_{j \neq i} \left(\frac{\partial(\mu_k - \mu_i)}{\partial C_j} \right) \nabla C_j \quad (12)$$

In a binary alloy, there is a unique alloy chemical potential gradient, $\nabla(\mu_B - \mu_A)$. The corresponding thermodynamic factor is defined as

$$\Phi = C_A C_B \frac{1}{k_B T} \frac{\partial(\mu_B - \mu_A)}{\partial C_B}. \quad (13)$$

Under irradiation, point defects are not at equilibrium and elimination reactions at point defect sinks give rise to gradients of point defect chemical potential. Therefore, in addition to the alloy chemical potential gradients, one accounts for point defect driving forces. If one ignores point defect clusters contributions to the point defect chemical potential, one may write vacancy and interstitial chemical potentials in terms of their equilibrium concentrations C_V^{eq} and C_I^{eq} respectively (dilute approximation):

$$\mu_V = k_B T \ln(C_V / C_V^{eq}) \quad \text{and} \quad \mu_I = k_B T \ln(C_I / C_I^{eq}), \quad (14)$$

leading to an expression of the associated driving force¹⁵

$$\frac{\nabla\mu_V}{k_B T} = \frac{1}{C_V} \nabla C_V - \frac{\xi_{VA}}{C_A} \nabla C_A, \quad \text{with} \quad \xi_{VA} = \frac{\partial \ln C_V^{eq}}{\partial \ln C_A}. \quad (15)$$

The interstitial driving force has the same form, except that letter V is replaced by letter I. Note, that the equilibrium point defect concentrations may vary with the local alloy composition and stress. Although the variation of the equilibrium vacancy concentration is expected to be mainly chemical, the change of the elastic forces due to a solute redistribution at sinks should not be ignored for the interstitials.¹⁵

Note that we introduce here a homogeneous formulation of the driving forces. The thermodynamic factors (i.e., partial derivatives of chemical potentials with respect to concentrations) depend on the local values but not on the spatial derivatives of the concentration field. This corresponds to a standard approximation in diffusion. However, it may become an issue when employed in continuous RIS models. Indeed, a typical RIS profile covers a few tens of nanometers. It may feature an oscillating behavior. Further, precipitation phenomena resulting from concentration spatial fluctuations may occur. Therefore, we expect local chemical potentials not only depend on the local value but also on the spatial derivatives of the concentration gradients. According to Cahn and Hilliard,⁷⁹ the free-energy model of a non-uniform system can be written as a volume integral of an energy density made up of a homogeneous term plus gradient energy terms proportional to the squares of concentration gradients. By ignoring the latter, a RIS diffusion equation cannot produce physical phenomena controlled by an interface energy. In particular, a RIS diffusion equation applied to the calculation of an equilibrium segregation profile of a structural defect gives rise to a flat profile. One cannot simulate an oscillating profile resulting from a competition between the thermodynamic driving force of the equilibrium segregation profile and flux coupling; and even less precipitation phenomena.

1.08.3.2.2 L-D relationship

A RIS model must be able to describe the reaction of a system to a gradient in the alloy composition but also to unusual driving forces, such as point defect concentration gradients. We write then the driving forces in terms of concentration gradients and introduce the diffusion coefficients as the constants of proportionality between fluxes and concentration gradients. They are functions of the L -coefficients and partial derivatives of chemical potentials with respect to the atomic and point defect concentrations.

A RIS model must also reproduce the atomic diffusion enhancement resulting from an increase of point defect concentration. If we ignore the thermodynamic and kinetic contributions of point defect clusters, the L -coefficients vary linearly with point defect concentrations. To highlight the role of point defect concentration, one introduces normalized L -coefficients, the so-called partial

diffusion coefficients, which are point defect concentration free.^{15,33,80} They depend on temperature and local alloy composition. First expressions of the partial diffusion coefficients in terms of the L -coefficients were from Howard and Lidiard⁸¹ for the vacancy in dilute alloys, from Barbu⁸² for the interstitial in dilute alloys, and from Wolfer¹⁵ in concentrated alloys. In the following, we use the formulation of Wolfer because the approximations made to calculate the L -coefficients and driving forces are clearly stated. In a binary alloy (AB), fluxes are separated into two contributions: the first one induced by the point defect concentration gradients, and the second one arising after the formation of chemical concentration gradients near the point defect sink¹⁵:

$$\begin{aligned} J_A &= C_A(d_{AV}\nabla C_V - d_{AI}\nabla C_I) - (d_{AV}^c C_V + d_{AI}^c C_I)\Phi\nabla C_A \\ J_V &= -(C_A d_{AV} + C_B d_{BV})\nabla C_V + C_V\Phi(d_{AV}^c - d_{BV}^c)\nabla C_A \\ J_I &= (C_A d_{AI} + C_B d_{BI})\nabla C_I + C_I\Phi(d_{AI}^c - d_{BI}^c)\nabla C_A \end{aligned} \quad (16)$$

where we define the partial diffusion coefficients in terms of the L -coefficients

$$\begin{aligned} d_{AV} &= \frac{L_{AA}^V + L_{AB}^V}{C_A C_V} \\ d_{AI} &= \frac{L_{AA}^I + L_{AB}^I}{C_A C_I} \\ d_{AV}^c &= \frac{L_{AA}^V}{C_A C_V} - \frac{L_{AB}^V}{C_B C_V} + d_{AV} \frac{1}{\Phi} \zeta_{VA} \\ d_{AI}^c &= \frac{L_{AA}^I}{C_A C_I} - \frac{L_{AB}^I}{C_B C_I} + d_{AI} \frac{1}{\Phi} \zeta_{IA} \end{aligned} \quad (17)$$

Note that the partial diffusion coefficients d have the same units as the L -coefficients or as a diffusion coefficient divided by the atomic volume: $\text{m}^{-1} \text{s}^{-1}$. ζ_{VA} is derived from the variation of the local equilibrium vacancy concentration with respect to the alloy local concentration (see Eq. (15)). Flux of B is deduced from the flux of A by exchanging the letters A and B. In a multicomponent alloy, equivalent kinetic equations are provided by the Perks's model.⁸³ In the latter, point defect concentrations are assumed to be independent of the chemical species concentrations: in other words, parameters ζ_{VA} and ζ_{VB} are set to zero. Most of the RIS models are derived from the Perks' model although they neglect the cross-coefficients and approximate the diagonal-coefficients by tracer diffusion coefficients.⁵ The flux of specie i is assumed to be independent of the concentration gradients of the other species. In doing so, not only the kinetic couplings, but also some of the thermodynamic couplings are ignored. Indeed, as shown in Eq. (12), a chemical potential gradient is a function of all the concentration gradients. However, relying on a CALPHAD database, Yang *et al.* have improved the Perk's model by adding crossed-thermodynamic terms controlling the dependence of atomic chemical potential on the full set of chemical species.⁶¹ Yet, one must emphasize, improving RIS models dedicated to concentrated alloys is not trivial, mainly because experimental diffusion data are lacking. An atomic flux results from a balance between the so-called IK effect, atomic fluxes induced by point defect concentration gradients (first term of the RHS of Eq. (16)), and the back diffusion and the Kirkendall (K) effect reacting against the formation of chemical concentration gradients at sinks produced by the IK effect (last term of LHS of Eq. (16)). Eq. (16) can be used in both dilute and concentrated alloys. Differences between models arise when one evaluates specific partial diffusion coefficients.

The first RIS model in dilute fcc alloys, designed by Johnson and Lam,⁸⁴ introduced an explicit variable for solute–point defect complexes. The same kind of approach has been used by Faulkner *et al.*,⁸⁵ although it has been shown to be incorrect in specific cases.^{82,86} The first RIS model derived from a rigorous estimation of fluxes, with a clear correspondence between the atomic jump frequencies and the L -coefficients, was devoted to fcc dilute alloys,⁸⁷ and then to bcc dilute alloys.⁸⁸

In concentrated alloys, due to the greater complexity and the lack of experimental data, atomic-based diffusion models of RIS are scarce. Most of them rely on TIP based diffusion equation, for which one uses empirical diffusion model as the Manning approximation to compute the diffusion coefficients.

1.08.3.2.3 Analytical solutions at steady state

In steady state, the time derivative of atomic and point defect concentrations are zero, in particular at the point defect sink. Therefore, the single flux coming from the bulk, which controls the atomic concentration (Eq. (3)) at the point defect sink is zero. Step by step from the sink plane, we demonstrate every flux of a given atomic species α is zero

$$J_\alpha = 0. \quad (18)$$

In case of a unique population of point defect sinks, setting to zero the time-derivatives of point defect concentrations in Eq. (2) leads to the relationship

$$J_V = J_I. \quad (19)$$

In a binary alloy, Eqs. (16), (18), and (19) yield the Wiedersich's relationship between concentration gradients:⁸⁰

$$\nabla C_A = \frac{C_A C_B d_{BI} d_{BV}}{(C_B d_{BI} D_A + C_A d_{AI} D_B)} \left(\frac{d_{AV}}{d_{BV}} - \frac{d_{AI}}{d_{BI}} \right) \nabla C_V, \quad (20)$$

where the intrinsic diffusion coefficient (normalized by the atomic volume) is equal to $D_\alpha = (d_{\alpha V}^c C_V + d_{\alpha I}^c C_I)\Phi$. The spatial extent of segregation coincides with the region of non vanishing defect gradients. Note that, in the original paper,⁸⁰ the partial and c-partial diffusion coefficients were taken to be equal. Such a simplification may change the amplitude of the RIS predictions. In a multicomponent alloy, not only the amplitude, but also the sign of RIS might be affected by this simplification.

Note that Eq. (20) only gives the relationship between ∇C_A and ∇C_V . In the special case where the elimination of point defects at sinks are dominant (i.e., when recombination is negligible) and for simple sink microstructures (thin films), one obtains analytical expressions of the point defect and solute concentration profiles, and of the total amount of solute segregation.²⁰ Main result is the solute segregation does not directly depend on the dose rate but only on the density of sinks.

1.08.4 Diffusion Equations: Application to Real Systems

It is often difficult to apply the previous models to a specific system due to the lack of diffusion experimental data. Below, we present a few experimental procedures to estimate both the driving forces and the L or D -coefficients. We introduce then the empirical models used to deduce the partial diffusion coefficients of a RIS model.

1.08.4.1 Diffusion Equation Parameters From Experiments

1.08.4.1.1 Thermodynamic databases

As shown in Eq. (13), we deduce the thermodynamic factor from a concentration derivative of the alloy chemical potential, which in turn is a first derivative of the Gibbs free energy per atom G of the alloy. Thus, we have

$$\Phi = C_A C_B \frac{1}{k_B T} \frac{\partial^2 G}{\partial C_B^2}. \quad (21)$$

When available, we may obtain an expression of the Gibbs free energy G from a thermodynamic database such as CALPHAD.⁸⁹ The latter builds free-energy composition functions of the alloy phases from thermodynamic measurements (specific heats, activities, etc.) and phase diagrams. These databases mostly fitted on high temperature experimental data are less valid at low temperature. There is a CALPHAD database for most of the binary alloys and a few ternary alloys, as for instance FeNiCr.⁹⁰ In case there is no thermodynamic database but a phase diagram, we may get a rough estimation of the Gibbs free energy from the regular solution model. In the specific case of a binary alloy $A_1\text{-}C B_C$ with a clustering tendency, the regular solution model of the Gibbs free energy per atom writes

$$G = 2k_B T_c C(1 - C) + k_B T C \ln(C) + k_B T(1 - C) \ln(1 - C), \quad (22)$$

where T_c is the critical temperature extracted from the phase diagram (corresponding to the maximum temperature of a miscibility gap in systems with a clustering tendency). From Eqs. (21) and (22), we deduce the concentration-dependent thermodynamic factor

$$\Phi = 1 - 4C(1 - C) \frac{T_c}{T}. \quad (23)$$

1.08.4.1.2 Experimental evaluation of the kinetic coefficients

The L -coefficients characterize the kinetic response of an alloy to a gradient of chemical potential. In practice, what we impose is a composition gradient. Thus, we switch to the $D - \nabla C$ writing of fluxes. Flux of a given species i follows then a generalized Fick's law

$$J_i = - \sum_j D_{ij} \frac{\nabla C_j}{\Omega}. \quad (24)$$

A diffusion experiment consists of measuring some of the terms of the diffusivity matrix D_{ij} . These terms cannot be determined one by one because at least two concentration gradients are involved in a diffusion experiment. Note, that the L -coefficients can be traced back only if the whole diffusivity matrix and the thermodynamic factors are known. Furthermore, most of the diffusion experiments are performed in thermal conditions and do not involve the interstitial diffusion mechanism.

In the following section, we present two thermal diffusion experiments. Then, we review the few diffusion experiments performed under irradiation. From these examples, we show how difficult it is to obtain the whole diffusivity matrix.

1.08.4.1.2.1 Interdiffusion experiments

In an interdiffusion experiment, a sample A (mostly composed of A atoms) is welded to a sample B (mostly composed of B atoms) and annealed at a temperature high enough to observe an evolution of the concentration profile. We assume vacancy is at local equilibrium and the gradient of vacancy chemical potential is zero. Therefore, according to Eqs. (4) and (16), the flux of component i in the reference crystal lattice is proportional to its concentration gradient:

$$J_i = - D_i \frac{\nabla C_i}{\Omega}, \quad (25)$$

where the so-called intrinsic diffusion coefficient D_i is a function of the phenomenological coefficients and the thermodynamic factor:

$$D_i = \Omega \left(\frac{L_{ii}^V}{C_i} - \frac{L_{ij}^V}{C_j} \right) \Phi. \quad (26)$$

An interdiffusion experiment consists of measuring the intrinsic diffusion coefficients as a function of local concentrations.

Interdiffusion experiments have been performed in austenitic and ferritic steels.³⁸ The determination of the intrinsic diffusion coefficients requires the measurement of the interdiffusion coefficient and of the Kirkendall speed for each composition.⁹¹ In general, an interdiffusion experiment provides the Kirkendall speed for one composition only, leading to a pair of intrinsic diffusion coefficients in a binary alloy. Therefore, few values of intrinsic diffusion coefficients have been recorded at high temperatures and on a limited range of the alloy composition. Nevertheless, even if a complete characterization of the intrinsic diffusion coefficients was possible, an interdiffusion experiment in an alloy of N_{sp} components gives access to N_{sp} intrinsic diffusion coefficients, which is not enough to trace back the full L -coefficient matrix comprising $N_{sp}(N_{sp} + 1)/2$ independent coefficients. In addition, interdiffusion is a kinetic response to a chemical driving force, which is very different from the kinetic response to a point defect concentration gradient. Anthony's diffusion experiment measures the flux coupling induced by a gradient of vacancy concentration, which is similar to what occurs under irradiation. Therefore, such an experiment should complement the interdiffusion experiments.

1.08.4.1.2.2 Anthony's experiment

Anthony set up a thermal diffusion experiment involving vacancies as a driving force^{21–25,28,92} in aluminum alloys. A gradient of vacancy concentration around a cavity is produced by a slow decrease of temperature. At the beginning of the experiment, the ratio between solute and vacancy fluxes is the following:

$$\frac{J_B}{J_V} = - \frac{L_{BB}^V + L_{AB}^V}{L_{AA}^V + L_{BB}^V + 2L_{AB}^V} \quad (27)$$

The volume of the cavity and the amount of solute segregation nearby yield a value for the flux ratio. Note, that an analysis relying on Eq. (27) neglects secondary fluxes induced by the formation of a segregation profile.

This experiment, combined with an interdiffusion annealing, could be a way to estimate the complete Onsager matrix. Unfortunately, the Anthony's experiment does not seem to be feasible in most alloys, especially in steels. In steels, vacancies do not form cavities, and solute segregation induced by quenched vacancies is not visible when the vacancy elimination is not concentrated on cavities. In addition, we suspect the large attraction between vacancy and impurities such as oxygen, nitrogen and carbon in iron hinders the formation of a vacancy chemical potential gradient at low temperature.⁹³

1.08.4.1.2.3 Diffusion under irradiation

In the 1970s, a few tracer diffusion experiments were performed under irradiation.⁹⁴ The main objective was to enhance diffusion by increasing point defect concentrations and thus accelerate diffusion at low temperature. Another motive was to characterize diffusion of the interstitials created by irradiation. In pure metals, some experiments were reliable enough to provide diffusion coefficient values at temperatures that were not accessible in thermal conditions.⁹⁴ The analysis of the same kind of experiments in alloys happened to be very difficult.⁹⁵ One of the major difficulties is to be able to decouple the effect of temperature and irradiation conditions on the microstructure from that on diffusion phenomena.

Another technique is to use irradiation to implant point defects at very low temperature. A slow annealing of the irradiated samples combined with electrical resistivity recovery measurement highlights several regimes of diffusion; at low temperature, interstitials with low migration energies diffuse alone, while at higher temperatures, vacancies and point defect clusters also diffuse. Temperatures at which a change of slope is observed yield effective migration energies of interstitials, vacancies, and point defect clusters.⁹⁶ *In situ* TEM observation of the growth kinetics of interstitial loops in a sample under electron irradiation is another method of determining the effective migration of interstitials.⁹⁷

1.08.4.2 L and D Coefficients From Empirical Models

In the previous section, we mentioned the difficulty in measuring the L -coefficients of an alloy, especially those involved in fluxes induced by point defect concentration gradients. Diffusion data are especially difficult to obtain for the interstitials. This is probably why most of the RIS models emphasize the effect of vacancy fluxes, with the interstitial contribution assumed to be neutral. We present the main approximations used to address the lack of experimental data and estimate the partial diffusion coefficients of the vacancy diffusion mechanism. Then, we discuss a few applications to the modeling of RIS in real alloys.

1.08.4.2.1 Manning-based model of concentrated alloys

First diffusion model for concentrated alloys under irradiation is due to Marwick.³³ He introduced the concept of radiation-enhanced diffusion by taking out vacancy concentration as a separate factor of the partial diffusion coefficients. We deduce the L -coefficient and the corresponding partial diffusion coefficients from a basic jump frequency model combined with the random lattice gas diffusion model of Manning.¹³ In this diffusion model, correlation effects are added as corrective terms of the basic

jump frequencies. The resulting fluxes³³ are similar to Eq. (16), except that the c-partial diffusion coefficients d_{AV}^c , d_{Al}^c are now equal to the partial diffusion coefficients d_{AV} , d_{Al} , which is equivalent to neglecting cross L -coefficients and the dependence of equilibrium point defect concentration on alloy composition. However, for the first time, one understands the segregation of the major element, Ni, in concentrated austenitic steel is the result of a competition between fast- and slow-diffusing major alloy components, Ni being the slow element and Cr the fast one.

After the relations of Manning,¹³ we obtain an approximation of the L -coefficients from the tracer diffusion coefficients D_i^V of element i :

$$L_{ij}^V = \frac{C_i}{\Omega} D_i^V \left[\delta_{ij} + \frac{1-f_0}{f_0} \frac{C_j D_j^V}{\sum_k C_k D_k^V} \right], \quad (28)$$

where D_i^V is the tracer diffusion coefficient of element i via the vacancy diffusion mechanism. The same relations can be written for the interstitial mechanism, even though in practice, we cannot measure the interstitial mediated tracer diffusion coefficients. The Manning relations systematically predict positive partial diffusion coefficients which are directly related to the tracer diffusion coefficients

$$d_{zV} = \frac{D_{z^*}^V}{\Omega C_V^{\text{eq}}}. \quad (29)$$

Establishing a link between partial diffusion coefficients and tracer diffusion coefficients can be very useful because we have a large database on tracer diffusion in model alloys of nuclear materials. However, we need to know the equilibrium concentration of vacancy in order to extract the normalized partial diffusion coefficient from Eq. (29). The latter may strongly depend on the alloy composition and temperature, and directly measuring it is very difficult.⁹³ Note the Manning relations imply relationships between the L -coefficients of an alloy, reducing to N_{sp} the number of independent L -coefficients. Such an approximation yields satisfying results on the RIS behavior of the major alloy components of concentrated alloys as for instance in austenitic steels.⁹⁸ In the latter, RIS is commonly thought of in terms of tracer diffusion coefficients. In binary alloys, one relies on the difference $[D_{A^*}^V/D_{B^*}^V - D_{A^*}^I/D_{B^*}^I]$ to predict the sign and the amplitude of RIS. However, such an approach may be not appropriate for the investigation of RIS of impurities in concentrated alloys. The constraint of positive L -coefficients implies no vacancy solute-drag whereas recent ab initio studies demonstrate the vacancy solute drag of impurities is very frequent, at least in dilute alloys.^{9,10} Since the solute drag mechanism is mainly due to the attraction between vacancy and solute atoms and the subsequent large variation of the vacancy exchange frequencies with respect to its distance to the solute, we expect the minor elements in concentrated alloys to be potentially exposed to the vacancy solute drag mechanism. We are not aware of a RIS model encompassing the solute drag mechanism in concentrated alloys.

Wiedersich *et al.*⁸⁰ added to Marwick's model a contribution of the interstitials. The global interstitial flux is described by Eq. (16), while preferential occupation of the dumbbell by a specie or two is deduced from the alloy concentration and the effective binding energies. Such a local equilibrium assumption lies in the hypothesis of very large interstitial jump frequencies compared to atomic jump frequencies. This kind of model yields an analytical expression of the ratio between the gradient of solute concentration and the gradient of defect concentration in the stationary state (see Eq. (20)). A little later, one has attempted to establish the relationship between elementary interstitial jumps, their change of composition and the resulting diffusion coefficients. From a microscopic description of the jump mechanism, one derives the kinetic equations associated with each dumbbell composition.^{99–102} Unlike previous models, there is no local equilibrium assumption, but correlation effects are neglected. Even more sophisticated interstitial diffusion model including the correlation effects had been developed,¹⁰³ but due to the lack of data for the interstitials, most of the concentrated alloy models have neglected the interstitial contribution to RIS.^{83,104,105}

1.08.4.2.2 Local concentration-dependent RIS model

Most of the RIS models assume thermodynamic factors equal to 1, although from the first modeling paper,¹⁵ the alloy thermodynamic factor was recognized to be varying with solute concentration. Similarly, the quantities ξ_{vx} and ζ_{lx} of the point defect driving forces depend on local concentration and stress field.^{15,106} The shape of RIS profile depends as well on the alloy thermodynamic factor.¹⁰⁷ Besides, Wolfer¹⁵ has demonstrated that RIS and its effect on the local thermodynamic factors affect the repartition between interstitial and vacancy fluxes and thereby, the swelling phenomena. The modification of sink bias might be due to several factors: the wind of point defect of a Kirkendall interdiffusion process induced by the formation of RIS, or the dependence of the point defect chemical potentials on local composition, including the elastic and chemical effects.

A local-concentration-dependent driving force is due to the local-concentration-dependent atomic jump frequencies. Following this idea, the modified inverse Kirkendall (MIK) model introduces partial diffusion coefficients of the form¹⁰⁴

$$d_{AV} = d_{AV}^0 \exp\left(\frac{-E_{AV}^m}{k_B T}\right) \quad (30)$$

The migration energy is written as the difference between the saddle-point energy and the stable state energy. It depends on local composition through pair interactions calculated from thermodynamic quantities such as cohesive energies, vacancy formation energies, and ordering energies. In fact, the present partial diffusion coefficients correspond to a mesoscopic quantity deduced from a coarse-grained averaging of the microscopic jump frequencies. In principle, not only the effective migration energies, but also the mesoscopic correlation coefficients and thermodynamic factors should depend on local concentrations.

Nevertheless, the thermodynamic factors, correlation coefficients, and the ζ_{Vz} factor are assumed to be those of the pure metal A. More recently, a new continuous model suggested a multi-frequency formulation of the concentration-dependent partial diffusion coefficients. Instead of averaging the sums of interaction bonds in the exponential argument, every jump frequency corresponding to a given configuration is considered with a configuration probability weight.¹⁸ Predictions of the model are compared with direct RIS Monte Carlo simulations that rely on the same atomic jump frequency models. In the two presented examples, the agreement is satisfactory. However, the thermodynamic factor and correlation coefficients are yet to be defined clearly.¹⁸

1.08.4.3 Diffusion Equation Parameters From Atomic Models

As we have seen in the previous section, a full experimental determination of the concentration and temperature dependent L -coefficients is almost impossible to achieve. An alternative approach is to compute the L -coefficients from the knowledge of atomic jump frequencies. First-principles methods are now able to provide accurate values of jump frequencies in a given local environment. In dilute alloys, only a few environments have to be considered and all the relevant jump frequencies can be estimated. In concentrated alloys, the environments are too many and *ab initio* calculations too time consuming: we still need to develop interatomic potentials and parameterized concentration-dependent model of the point defect jump frequencies. Once we have built the jump frequency model, the L - or D -coefficients can be estimated from diffusion theories or from atomistic simulations.

1.08.4.3.1 Jump frequencies

In the framework of the transition state theory,¹⁰⁸ the exchange frequency between a vacancy V and a neighboring atom A is given by:

$$\Gamma_{AV} = v_{AV} \exp\left(-\frac{\Delta G_{AV}^{mig}}{k_B T}\right) = v_{AV} \exp\left(+\frac{\Delta S_{AV}^{mig}}{k_B}\right) \exp\left(-\frac{\Delta H_{AV}^{mig}}{k_B T}\right) \quad (31)$$

when ΔH_{AV}^{mig} is significantly greater than thermal fluctuations $k_B T$. ΔG_{AV}^{mig} , ΔH_{AV}^{mig} and ΔS_{AV}^{mig} are respectively the Gibbs free energy, enthalpy and entropy of migration and v_{AV} is the attempt frequency. ΔH_{AV}^{mig} corresponds to the increase of the total enthalpy when A goes from its initial site on the crystal lattice to the saddle point between its initial and final position. A similar expression holds for self-interstitial jumps. The description of these jump frequencies, and of their dependence on the local atomic configuration, encompasses all the information on the thermodynamic and diffusion properties of the system.

1.08.4.3.1.1 *Ab initio* calculations

Since the development of the density functional theory (DFT), first-principle methods have dramatically improved our knowledge of point defect and diffusion properties in metals.^{109–114} They provide a reliable way to compute the formation and binding energies of defects, their equilibrium configuration and migration barriers, the influence of the local atomic configuration in alloys, etc. The DFT studies on self-interstitial properties – for which few experimental data are available – are of particular interest. They have shown e.g., that the stable configuration is the [110] dumbbell in α -iron,^{115,116} and the [100] dumbbell in most fcc alloys.^{113,114} More recently, one has demonstrated a systematic investigation of the thermodynamic and kinetic properties of dumbbells in dilute Fe-based alloys, including the vibrational entropy contributions, is now feasible.¹²

However, some limitations remain. Migration enthalpies are usually computed at 0K by the drag or by the nudged elastic band methods.¹¹⁷ Migration entropies can be computed by phonon calculations within a harmonic or quasi-harmonic approximation, but these calculations are especially expensive and since the jump frequency is less sensitive to the variation of ΔS_{AV}^{mig} than to the variation of ΔH_{AV}^{mig} , they are frequently dispensed with (a constant ΔS_{AV}^{mig} is assumed and integrated in the attempt frequency). More generally, the determination of finite temperature effects is still incomplete. Direct calculations of ΔG_{AV}^{mig} beyond the harmonic approximation are scarce and rarely applied to RIS problems. The modeling of magnetic transitions affecting both point defect and diffusion properties (especially in iron-based alloys), is mostly phenomenological.^{111,118,119} Above all, DFT methods are still too time consuming to allow either direct simulations of the atomic displacement fluctuations, ‘on-the-fly’ calculations of the migration barriers, or their prior calculation and tabulation for all the possible local configurations (whose number increases very rapidly with the range of interactions and the number of chemical elements). One recent exception is the measurement of self-interstitial tracer diffusion coefficients in Ni–Cr alloy, by *ab initio* molecular dynamics.¹²⁰ An alternative approach is to develop approximate models of local-concentration dependent jump frequencies for concentrated alloys, whose parameters are fitted to reproduce experimental data and/or *ab initio* calculations.

1.08.4.3.1.2 Interatomic potentials

Empirical or semi-empirical interatomic potentials, usually developed for molecular dynamics simulations, can be used for the modeling of RIS, but one must address several issues:

- To get a reliable description of an alloy, the interatomic potential has to be adjusted on properties that control the flux coupling of point defects and chemical elements. A complete fitting procedure would be very tedious and, to our knowledge, has never been achieved for a given system.

- Direct measurement of the L -coefficients by MD are very demanding in CPU-time. Up to now, one is able to measure tracer diffusion coefficients mediated by an SIA diffusion mechanism only. For example, Terentyev *et al.*^{121,122} have measured this way the tracer diffusion coefficient of Cr and Fe in bcc Fe–Cr alloys with SIA. They have found that the preferential transport of Cr by $\langle 110 \rangle$ dumbbells should tend to an enrichment of Cr near point defect sinks in ferritic steels.
- At lower temperature, MD can be combined to a static calculation of jump frequencies in terms of attempt frequencies and migration barriers. The direct calculation of migration barriers with an interatomic potential, even though much simpler than DFT calculations, is still quite time consuming. Full calculations of vacancy migration barriers have indeed been implemented in Monte Carlo simulations,¹²³ using massively parallel calculation methods, but they are still limited to relatively small systems and short times, for example, for the study of diffusion properties rather than microstructure evolution. It is possible to simplify the calculation of the jump frequency, for example, by not performing the full calculation of the attempt frequencies (their impact on the jump frequency must be less critical than that of the migration barriers involved in the exponential term) or by omitting the relaxation of the saddle-point position.¹²⁴ Malerba *et al.* have recently proposed another method where the point defect migration barriers of an interatomic potential are exactly computed for a small subset of local configurations, the other ones being extrapolated using artificial intelligence techniques. This has been successfully achieved for the modeling of diffusion of vacancies in iron–copper alloys.^{125,126} Such techniques have not yet been used to model RIS, but this could change in the future.

1.08.4.3.1.3 Broken-bond models

Because of these difficulties, simulations of diffusive phase transformation kinetics are commonly based on various broken-bond models, in the framework of rigid lattice approximations.^{5,127} The total energy of the system is considered to be a sum of constant pair interaction energies, for example, $\varepsilon_{AB}^{(n)}$ between A and B atoms located on n th nn sites. Interactions between atoms and point defects can also be used to provide a better description of their formation energies and interactions with solute atoms, and other defects. Concentration dependent pair interactions can be used to reproduced asymmetrical thermodynamic properties.¹⁴

Various approximations are used to compute the migration barriers: a common one¹²⁸ is writing the saddle-point energy of the system as the mean energy between the initial energy E_i and final energy E_f , plus a constant contribution Q (which can depend on the jumping atom, A or B). The migration barrier for an A–V exchange is then:

$$\Delta E_{AV}^{mig} = \frac{E_f - E_i}{2} + Q_A \quad (32)$$

where $E_f - E_i$ corresponds to the balance of bonds destroyed and created during the exchange.

Another solution is to explicitly consider the interaction energy e_{AV}^{SP} of the jumping atom A with the system, when it is at the saddle point:

$$\Delta E_{AV}^{mig} = e_{AV}^{SP} - \sum_{i,n} \varepsilon_{Ai}^{(n)} - \sum_{j,n} \varepsilon_{Vj}^{(n)} \quad (33)$$

e_{AV}^{SP} itself can be written as a sum of interactions between A and the neighbors of the saddle point.^{60,129,130}

We easily extend both approximations to the interstitial diffusion mechanisms, as long as we have experimental and/or *ab initio* data to rely on for the fitting of the interaction energies. The first approximate model of the migration energy has the drawback of imposing a linear dependence between the barrier and the difference between the initial and final energies, which is not justified and has been found to be unfulfilled in the very few cases where it has been checked using empirical potentials.¹²³ The second model should better take into account the effect of the local configuration and, according to the theory of activated processes, does not impose a dependence of the barrier on the final state. However, a model of pair interactions on a rigid lattice does not give a very precise description of the energetic landscape in a metallic solid solution, so, the choice of approximations may not be crucial. Taking into account many-body interactions (fitted to *ab initio* calculations, using cluster expansion methods) could improve the description of migration barriers, but would significantly increase the simulation CPU time.^{131,132}

1.08.4.3.2 Calculation of the phenomenological coefficients

Given an atomic jump frequency model, transitions of the alloy configurations are described by a master equation. With one point defect in the system, a Monte Carlo simulation produces a trajectory of the alloy in the configuration space. The L -coefficients can be obtained from a Monte Carlo simulation. Measurements are performed on an equilibrated system using the generalized Einstein formula of Allnatt:¹³³

$$L_{ij} = \frac{1}{\Omega} \frac{\langle \Delta R_i \Delta R_j \rangle}{6t} \quad (34)$$

where ΔR_i is the displacement of the species i during time t . This numerical approach has proved its efficiency; however it is limited to short ranges of composition and temperature. Simulations become rapidly unworkable when, for example, binding energies between interstitials and neighboring atoms are significant.^{134,135} The simulated trajectory can be trapped in some configurations due to the correlation effects of the diffusion mechanism; after a jump, an atom has a finite probability to exchange again with the same point defect and cancel its first jump. The escape probability of the point defect from an atom decreases with the binding energy between the two species.

In the limited case of dilute alloys, in which a few point defect jump frequencies are involved, it is possible to consider all the vacancy paths and deduce semi-analytical expressions for the L -coefficients. In concentrated alloys, mean field diffusion models of concentrated alloys yield approximate expressions of the transport coefficients.

1.08.4.3.2.1 Dilute alloys

The point defect jump frequencies to be considered are those that are far from the solute, those leaving the solute, those arriving at a nn site of the solute, and those jumping from nn to nn sites of the solute. We may apprehend a solute diffusivity as a series in which the successive terms include longer and longer looping paths of the point defect from the solute. A first shell diffusion model accounts for solute-PD interactions between 1NN sites only, so that the set of exchange frequencies to be considered is small (five in fcc,⁷ four in bcc⁷ and eight in hcp alloys¹³⁶ for the vacancy mechanism).

In the case of the vacancy diffusion mechanism, before the development of first-principles calculation methods, 1nn vacancy-solute interactions only were usually considered and jump frequency ratios could be estimated from a few experimental diffusion coefficients (three for fcc and two for bcc). Data were calculated from the solvent and solute tracer diffusivities, the linear enhancement of self-diffusion with solute concentration, or the electro-migration enhancement factors of tracer atoms in an electric field.¹³⁷ Some examples of jump frequency ratios fitted to experimental data are displayed in published tables.¹³⁸ Currently, the most widely used approach is using first-principles calculations to calculate not only the vacancy, but also the interstitial jump frequencies.^{12,113,114}

Exact analytical expressions of the corresponding Onsager coefficients for these multi-frequency diffusion models can be obtained.¹³⁹ A very large database of self and impurity diffusion coefficients,¹⁴⁰ and flux couplings,¹¹ has recently been automatically generated from high-throughput DFT calculations,¹⁴⁰ for more than 230 dilute solute diffusion systems in Mg, Al, Cu, Ni, Pd, and Pt host lattices. The good agreement with experimental data confirms the validity of the approach. Thanks to the recent development of more sophisticated diffusion models, which account for the variation of PD frequencies beyond the 1NN interaction shell, one may investigate not only tracer diffusion but also flux couplings. The latter crucially depend on the variation of the PD frequencies with local environment. To compute them, we may use the code KineCluE (for Kinetic Cluster Expansion)⁸ based on the self-consistent mean field (SCMF) theory,^{9,109,111,141,142} and for the specific case of vacancy-mediated diffusion, the code Onsager based on mixed SCMF-Green function formalism.¹⁰ High throughput DFT approaches of low temperature flux couplings in dilute alloys have now become feasible.^{9,10} In the high temperature regime, L_{AV} and L_{BV} are negative, and tracer diffusion and solute-PD flux coupling mainly depend on the exchange frequency of a vacancy with an atom of the matrix far from the solute and the exchange frequency of a vacancy with the solute atom. Consequently, the knowledge of the self-diffusion and solute diffusion coefficients usually provides a good estimation of L_{AV} and L_{BV} and of the vacancy-solute flux coupling. This corresponds to the traditional interpretation of RIS in terms of inverse Kirkendall effect: the elimination of vacancies leads to an enrichment of solute B at sinks when $d_{BV} < d_{AV}$ and to a depletion when $d_{AV} < d_{BV}$. However, at low temperature, large variations of the vacancy exchange frequencies with respect to its proximity with solute atoms may yield sequences of strongly correlated successive jumps of the vacancy. These correlations are the main contributors to the atomic-species off-diagonal Onsager coefficients and determine the sign of flux coupling between PDs and atomic species. In its course toward a point defect sink, the vacancy turns around a solute atom by jumping from a neighboring site to another, then exchanges with it; and this way drags solute atoms toward the sink. This solute drag mechanism is associated with a positive flux coupling ($L_{BV} > 0$). It results from a subtle game of PD-solute binding energies and kinetic correlations and cannot be predicted from high temperature diffusion experiments. Studies dedicated to Fe-X dilute alloys report such a systematic behavior with negative L_{XV} at high temperatures and positive L_{XV} at low temperature.⁹ The Fe-Cr system is a special case where the binding energy between Cr and V is especially weak. Consequently, the drag effect occurs at lower temperature than in other Fe-based binary alloy: below 300K, where no RIS is expected.

The amplitude as well as the sign of RIS depend on both vacancy and interstitial contributions. Very recently, one has performed a similar systematic analysis of the solute-interstitial coupling in Fe-based alloys.¹² Preferential solute transport by dumbbells occurs for P, Mn, and for Cr to a lesser extent.

The use of symmetry operations for the analysis of the PD transitions and the computation of the Onsager coefficients have recently widen the investigations to more complex systems.^{139,143,144} Let us mention the study of solute drag in strained systems,¹⁰⁹ in ternary alloys,¹⁴³ and in the hexagonal close packed magnesium crystal.¹⁰

Finally, it is worth noting that in systems submitted to irradiation, the solute-point defect clusters are not necessarily in local equilibrium with each other. In that case, we cannot use the TIP formulation of fluxes. A recent extension of the SCMF theory splitting the Onsager matrix of L -coefficients into intrinsic matrices of cluster L -coefficients, should be particularly useful for the analysis of flux coupling associated with a non-equilibrium cluster distribution.^{8,145} As well the forced relocation of atoms within a displacement cascade may be treated as an atomic diffusion mechanism, although it does not satisfy the detailed balance, i.e., it does not drive the system toward the equilibrium state. Recent theoretical developments implemented into the code KineCluE allow to take into account the effect of a forced relocation mechanism on steady state short range order and on flux couplings.¹⁴⁶

1.08.4.3.2.2 Concentrated alloys

The first diffusion models devoted to concentrated alloys rely on a very basic description of the diffusion mechanism. One equates the alloy to a random lattice gas model where atoms do not interact and where vacancies jump at a frequency that depends only on the species they exchange with (two frequencies in a binary alloy). Using complex arguments, Manning¹³ could express the correlation

factors as a function of the few jump frequencies. The approach was extended to the interstitial diffusion mechanism.¹⁰³ Such diffusion models, considering a limited number of jump frequencies, already make spectacular correlation effects appear possible, as for example percolation phenomena when the host atoms are immobile.^{13,147,148} However, they do not account for the effect of short-range order on the L -coefficients although one knows that RIS behavior is often explained by means of a competition between binding energies of point defects with atomic species, especially in dilute alloys. One made a few attempts to incorporate short-range order in a Manning-type formulation of the phenomenological coefficients, but coherency with thermodynamics was not guaranteed.^{149,150}

The current diffusion models for concentrated alloys, including short-range order, are based on either the path probability method (PPM)^{151–153} or the SCMF theory.^{148,154–157} Both mean-field methods start from an atomic diffusion model and a microscopic master equation. While the PPM considers transition variables, which are deduced from a minimization of a pseudo free-energy functional associated to the kinetic path, the SCMF theory introduces an effective Hamiltonian to represent the non-equilibrium correction to the distribution function probability and calculates the effective interactions by imposing a self-consistent constraint on the kinetic equations of the distribution function moments. Diffusion models built from the PPM were developed in bcc solid solution and ordered alloys, using nn pair transition variables, which is equivalent to considering nn effective interactions, and a statistical pair approximation for the equilibrium averages within the SCMF theory.^{151,152} The SCMF theory extended the approach to fcc alloys^{13,147,148} and provided a model of the composition effect on solute drag by vacancies.¹⁵⁵ The interstitial diffusion mechanism was also tackled.^{135,158,159} For the first time, an interstitial diffusion model including short-range order was proposed in bcc concentrated alloys.¹⁵⁸ It was shown that the usual value of 1 eV for the binding energy of an interstitial with a neighboring solute atom leads to very strong effects on the average interstitial jump frequency and, therefore, on the L -coefficients.

1.08.4.4 Comparison With Experiment

RIS almost systematically takes place upon electron, neutron and ion irradiation, even at very low radiation dose. In many cases, we cannot achieve a full quantitative comparison between the RIS models and experimental observations, mostly because samples are made from commercial alloys containing at least five different chemical species plus impurities while models simulate RIS in binary and ternary model alloys. In addition, we generally simulate RIS as an isolated phenomenon while its amplitude and even its sign (in ferritic steels) depends on the full radiation-induced microstructure. We present a few examples of investigations, first in dilute alloys, then in concentrated alloys. We close the section on the main challenges we still have to tackle.

1.08.4.4.1 Dilute alloys

RIS measurements in dilute alloys are less numerous than in concentrated alloys because the required grain boundary concentrations are usually smaller. However, some of the first RIS observations concerned the segregation of a dilute element, Si, in austenitic steels. In this specific case, observations were easy because the RIS of Si goes along with the precipitation of Ni_3Si .

The first mechanism proposed to explain the observed solute segregation was the diffusion of solute–point defect complexes towards sinks.⁸⁴ Since then, more rigorous models that rely on the linear response theory have been established and applied to the RIS description of Mn and P in nickel⁸⁷ and P in ferritic steels.^{86,88} Although good precision of the microscopic parameters was still missing, the formulation of the kinetic equations was general enough to be used almost without modification.⁸⁴ Recent *ab initio* calculations not only provided accurate atomic jump frequencies of P in Fe,^{115,160} but they also called into question the jump interstitial diffusion mechanism that had to be considered.¹⁶⁰ Indeed, the octahedral and the (110) mixed dumbbell configurations have almost the same stability and migration enthalpies. The resulting effective diffusion energy estimated by the transport model was found to be smaller than the self-interstitial atom migration enthalpy, confirming the classical statement that a solute atom with a negative size effect tends to segregate at the grain boundary. However, as emphasized in Meslin *et al.*,¹⁶⁰ the current interpretation of the interstitial contribution to RIS in terms of size effects is certainly oversimplified. A very large *ab initio* value of 1.05 eV for the binding energy between a mixed dumbbell and a substitutional P atom may lead to a large activation energy for P interstitials and a drastic reduction of P segregation predictions.¹⁶⁰ To consider this new blocking configuration with two P atoms, a concentrated alloy diffusion model including short-range-order effects is required.

A systematic RIS observation of Cr and minor elements in four martensitic-ferritic alloys irradiated with 2.0 MeV protons over a range of doses (1–10 dpa) and temperatures (300–700°C) has been performed. A positive and a bell-shaped temperature dependence of RIS in minor elements Si, Ni, and Cu have been observed.⁶⁶ DFT studies of dilute Fe-base binary alloys predict a drag by vacancy of these minor elements and a positive RIS at temperatures below 700°C.^{111,161,162} These theoretical results although not performed on industrial multi-component alloys are consistent with the experimental observations.

1.08.4.4.2 Austenitic steels

Most of the RIS models for concentrated alloys were devoted to the ternary Fe–Cr–Ni system, which is a model alloy of austenitic steels. The diffusion ratios used in the fitting process are the ones extracted from the tracer diffusion coefficients measured by Rothman *et al.* (referenced in Perks and Murphy⁸³ and Allen and Was¹⁰⁴) and more recently the ones collected by Jonsson¹⁶³ for the model of Yank *et al.*⁶¹ In most of the studies, the input parameters are taken from Perks model.⁸³ The more recent MIK model, which was initially based on the Perks model, used CALPHAD database to fit its concentration-dependent migration energy model,¹⁰⁴ and the thermodynamic factors.⁶¹ A significant improvement in the predictive capabilities of RIS modeling was achieved

after a systematic comparison with RIS observations. RIS data which have been used for benchmarking the models are mainly Auger spectroscopy measurements of RIS in Fe–Cr–Ni model alloys as a function of temperature, nominal composition, and irradiation dose,¹⁰⁴ and scanning transmission electron microscopy-energy-dispersive x-ray spectroscopy profiles of neutron irradiated stainless steels with respect to the radiation dose.⁶¹

Every model reproduces the correct tendencies of RIS in austenitic steels: a depletion of Cr and an enrichment of Ni near a grain boundary. Therefore, whenever the binding energies of point defects with atoms are not so strong and the ratios between the tracer diffusion coefficients of the major elements are large enough (larger than 2–3), a rough estimation of the partial diffusion coefficients from tracer diffusion coefficients seems to be sufficient to reproduce the main tendencies.

As explained above, one often neglects interstitial contribution to RIS due to the lack of diffusion data. However, Stepanov *et al.*¹⁶⁴ observed an electron-irradiated foil at a temperature low enough so that only interstitials contributed to the RIS. Segregation profiles were similar to the typical ones observed at higher temperature. Parameters of the interstitial diffusion model were estimated in such a way that the experimental RIS was reproduced. The migration energy of interstitials was assumed to be equal to 0.2 eV, which is quite low in comparison with the effective migration energy deduced from recovery resistivity measurements.¹⁶⁵ More recently, parametrization of the interstitial diffusion model was achieved by relying on DFT formation energy of the interstitials depending on their composition and experimental observation of RIS profiles.⁶¹ DFT calculations tend to suggest Cr-rich interstitial is more stable than Ni-rich interstitial in the Fe-rich Fe–Cr–Ni alloys.¹⁶⁶ Main conclusion of this new RIS model is the dominant diffusion mechanism is the vacancy mechanism for Fe and Cr, while for Ni, the diffusion swings from a V-dominant mechanism to a SIA dominant mechanism.

RIS in austenitic steels is commonly used as a benchmark to assess the representativity of neutron irradiation by ion and electron irradiation. By relying on a point defect rate theory model, we may estimate the flux-temperature transition lines between the three main kinetic regimes, the thermal one (back diffusion), the one dominated by the elimination at sinks (RIS) and the one dominated by recombination (see Fig. 3). Based on the same rate theory model, Mansur has derived analytical expressions of the dose rate and dose dependence of the amount of absorbed point defects at sinks.¹⁶⁷ We may then deduce the shift of temperature, which is required to keep the same amount of absorbed point defects after a change of the dose rate. Assuming RIS is mainly controlled by the latter, the group of Was applies the temperature shift method to reproduce the RIS behavior of neutron-irradiated austenitic steels by means of self-ion irradiation.¹⁶⁸ Nevertheless Eq. (20) shows RIS is not exactly sensitive to the amount of absorbed point defects at sinks but to the gradient of a point defect concentration divided by its concentration. In the elimination-at-sink dominant regime, the resulting RIS should not vary with the dose rate, unless the point defect sink microstructure has sensibly changed.²⁰

The attempts of the first MIK model to reproduce the characteristic ‘W-shaped’ Cr profiles observed at intermediate doses were not conclusive: transitory profiles disappeared after a dose of 0.001 dpa, while the experimental threshold value was around 1 dpa.⁴⁰ The more recent MIK model adding the interstitial contribution to RIS and thermodynamic factors different from one, highlight a ‘W-shaped’ Fe and Ni profiles at 0.1 dpa.⁶¹ A study based on a lattice rate theory pointed out that an oscillating profile was the signature of a local equilibrium established between the surface plane and the next plane.¹⁶⁹ This kind of mean-field model predicts that the local enrichment of Cr at a surface persists at larger irradiation dose (0.1 dpa), though not as large as the experimental value.

The role of impurities as point defect traps has been explored since the 1970s.¹⁷⁰ In those models, impurities do not contribute to fluxes but to the sink population as immobile sinks with an attachment parameter depending on an impurity–point defect binding energy.¹⁷¹ Other models account for immobile vacancy traps by renormalizing the recombination coefficient with a vacancy–impurity binding energy.¹⁶⁴ Whether by vacancy or by interstitial trapping, the result is a recombination enhancement and a decrease of point defect concentrations, leading to a reduction of RIS and swelling. Hackett *et al.*¹⁷¹ estimated some binding energies between a vacancy and impurities, such as Pt, Ti, Zr, and Hf in fcc Fe, using *ab initio* calculations. The energy calculations seem to have been performed without relaxing the structure, probably because fcc Fe is not stable at 0 K. Although the absolute values of the binding energies should be used with caution, one can expect the strong difference between the binding energies of a vacancy with Zr (1.08 eV) and Hf (0.71 eV) to persist after a relaxation of atomic positions. In a more rigorous way, the trapping of dumbbells could be modeled using the high migration energy associated with dumbbells bound to an impurity.⁶⁰ Such a model would allow the impurity to migrate and change the sink density with dose. The same model could explain the recent experimental results observing that, after a few dpa, RIS of the major elements starts again.¹⁷¹

RIS in austenitic steels was observed to be strongly affected by the nature of the grain boundary, that is, by the misorientation angle and the Σ value.⁴⁴ One explains differences between the observed RIS by means of the so-called grain boundary efficiency, which is taken into account in a modified rate equation model.^{44,99,105,172} Calculations of vacancy formation energies at different grain boundaries, for example, in nickel in which atomic interactions are described by an embedded atom method, have been used to model sink strength as a function of the misorientation angle. The resulting RIS predictions around several tilt grain boundaries were found to be in good agreement with RIS data.¹⁷² Grain boundary displacement and its effect on RIS were considered too. Grain boundary displacement was explained by an atomic rearrangement process due to recombination of excess point defects at the interface. New kinetic equations including an atomic rearrangement process after recombination of point defects at the interface predict asymmetrical concentration profiles, in agreement with experiments.¹⁰⁵

1.08.4.4.3 Ferritic steels

More recently, a RIS model based upon the framework of the Perk’s model has been parameterized for ferritic steels.¹⁷³ It relies on the very strong assumption that every partial diffusion coefficient is defined by a single effective migration energy which value is selected

among a set of DFT migration barriers computed in dilute configurations. Despite this approximation, the model is able to take into account the competition between the RIS contributions of self-interstitials and vacancies. The former prevails at moderate temperatures, leading to Cr enrichment, the latter at higher temperatures, leading to Cr depletion – in good agreement with the experimental results. The decreasing amount of Cr enrichment with increasing bulk Cr composition is also reproduced.^{59,66,173}

1.08.4.5 Challenges of the RIS Continuous Models

Ab initio calculations have become a very powerful tool for RIS simulations. They have been shown to be able to provide not only the variation of the atomic jump frequencies with local concentration,¹⁷⁴ but also new diffusion mechanisms.¹⁶⁰ From a precise knowledge of the atomic jump frequencies and the recent development of diffusion models,¹⁵⁵ a quantitative description of flux couplings is now feasible in dilute alloys. In concentrated alloys, it requires first a development of a local concentration dependent jump frequency model of both vacancy and self-interstitial, and second the computation of the partial diffusion coefficients and the thermodynamic factors. Whereas both steps have been achieved in Fe–Cr alloys,¹⁰⁷ the first step is still an on-going work for austenitic steels.

Radiation induced precipitation following RIS is a common phenomenon. Including precipitation mechanisms in a continuous approach is not straightforward because it implies a quantitative modeling of the spatial concentration fluctuations. Adding fluctuations within a deterministic approach requires a proper coarse graining method.^{175,176} By introducing physical concentration fluctuations, they naturally produce thermodynamic and kinetic properties, which do not depend on the discretization of the solid. Even though, they are able to simulate precipitation phenomena in addition to flux coupling, they rely on a local equilibrium assumption within the cell, which prevent them to tackle nucleation phenomena featuring a critical nucleus smaller than the discretized cell.

Although it has been suggested since the first publications³³ that the vacancy flux that counters the built-up gradients of RIS could slow down the void swelling, the change of microstructure and its coupling with RIS have almost never been modeled. Only very recently, phase field methods have tackled the kinetics of a concentration field and its interaction with a cavity population or a single dislocation loop. The development of a simulation tool able to predict the mutual interaction between the point defect microstructure and the flux coupling is still challenging.

1.08.5 Simulation of RIS From Atomic Jumps

There are two main strategies to predict RIS from the atomic jumps. The first one aims at computing the macroscopic parameters of a RIS continuous model from the atomic scale, as explained above. The alternative approach is to perform full atomistic simulations of the segregation, relying on the atomic-scale description of the point defect jump mechanisms. One advantage of the method is that it does not require a diffusion model, based on specific approximations, to make the link between the jump frequency and the transport coefficient.

Molecular dynamics methods are much too slow – by several orders of magnitude – for microstructure evolution governed by thermally activated migration of point defects. Atomic lattice rate models or Atomic Kinetic Monte Carlo (AKMC) methods are therefore the appropriate technique to simulate nanoscale phenomena such as RIS. Phase-Field models are especially useful to model large systems and the effect of elastic interactions, but their kinetic properties must be first carefully parametrized using another method, such as AKMC simulations. The difficulties are the same as for the modeling of diffusive phase transformations (such as precipitation or ordering during thermal aging), although one has to deal with the non-conservation of the point defect concentration resulting from several reactions: the point defect production, recombination and elimination at point defect sinks of the microstructure. Besides, the self-interstitial jump mechanisms are usually more complex than the vacancy ones.¹²⁷

Validity of both approaches lies in the physical foundations of an on-lattice modeling of the point defect jump frequencies and the interaction of point defects with the ever-changing microstructure.

1.08.5.1 Creation and Elimination of Point Defects

Because RIS is due to fluxes of excess point defects, modeling must take into account their creation and elimination mechanisms. For example, it must reproduce the ratio between vacancy and interstitial concentrations that controls the respective weights of annihilation by recombination or elimination at sinks. The situation from this point of view is very different from phase transformations during thermal aging, where usually only vacancy diffusion occurs, and simulations can be performed with nonphysical point defect concentrations and a correction of the timescale.^{129,174,177}

During electron or light ion irradiation, defects are homogeneously created in the material, with a frequency directly given by the radiation flux (in dpa s⁻¹), a condition that is easily modeled in continuous models, mean-field models,^{60,178} or Monte Carlo simulations.^{17,161} The formation of defects in displacement cascades during irradiation by heavy particles can also be introduced in kinetic models, using the point defect distributions computed by molecular dynamics.¹⁷⁹

The annihilation mechanisms at sinks such as surfaces or grain boundaries are, for the time being, simulated using very simple approximations (perfects sinks, no formation/annihilation of kinks on dislocations, or steps on surfaces). This should not affect

the basic coupling between diffusion fluxes, but the long-term evolution of the sink microstructure – which will eventually have an impact on the chemical distribution – is not taken into account. Nevertheless, one may investigate the impact of the grain size and/or point defect sink density on RIS by varying the distance between parallel perfect grain-boundaries.²⁰ Besides, one may introduce in an empirical way the impact of point defect clusters contributing to the point defect sink population, and in turn to RIS. It has been shown recently, a RIS simulation method combined with a mean field cluster dynamics method simulating the evolution of the point defect clusters, yields realistic simulation of radiation-enhanced precipitation in Fe–Cr.¹⁸⁰

Thermally activated point defect formation mechanisms that operate during thermal aging, are taken into account in continuous simulation methods.^{15,107,169,178} Simulations that do not include the thermal production are then valid only at sufficiently low temperatures, when defect concentrations under irradiation are much larger than equilibrium ones.

1.08.5.2 Lattice Rate Models

The first atomic scale models were mean field lattice rate models. They included two thermally activated jump frequencies, one for the vacancy and the other for the self-interstitial, first as a direct interstitial diffusion mechanism¹⁷⁸ and later on as a dumbbell diffusion mechanism.⁶⁰ Although modeled at the atomic scale, one computes the jump frequency as a function of a mean field migration energy depending on the mean field occupation of surrounding lattice sites both at the saddle point and at the equilibrium state. In general, one relies on a broken bond model to express the variation of migration energies in function of the local composition. At that time, DFT studies were scarce. Hence, energy interactions were fitted on thermodynamic data including the phase diagram, surface and vacancy formation energy. Such a model is able to capture the local driving forces in an approximate way but its description of the local dynamics of point defects is limited. In principle, atomic mean field kinetic equation do not encompass the kinetic correlations. In particular, they should be able to model a change of kinetic regime with respect to the variation of the local solute concentration, as for instance a transition from a solute drag flux coupling to a solute-vacancy negative flux coupling. Nevertheless, in systems like austenitic steels, vacancy flux coupling is always negative. Thus, one may adopt an empirical approach and parameterize the vacancy jump frequencies to reproduce the macroscopic diffusion properties of the alloy.⁶⁰ As for the self-interstitial parameters, we adjust them to reproduce the effective migration energies extracted from resistivity recovery measurements.¹⁶⁵ The resulting local-concentration-dependent jump frequencies describe both the kinetics of thermal alloys toward equilibrium and irradiation-induced surface segregation in concentrated alloys. Further, describing diffusion at the atomic scale yields atomic details on the thermodynamic and kinetic properties of point defects and atoms close to the point defect sinks. For instance, one may study the kinetics of establishment of a local equilibrium near a free surface, acting as a source and sink for point defects. We model it by introducing above the surface, a buffer plane almost full of vacancies. Hence, this mean field method naturally takes into account the transition of a segregation profile from the thermal equilibrium one to the RIS one. Application to model alloys of austenitic steels highlights the ‘W-shaped’ profile is a transient state from the pre-irradiated enrichment to the strong radiation-induced depletion of Cr.¹⁸¹ Therefore, both thermodynamic properties and the ratio of the transport coefficients between Fe, Cr, and Ni determines an RIS profile (Fig. 9). Note the predicted W-profile is not as wide as the experimental one.

What needs to be improved is the interstitial diffusion model. The lack of experimental and *ab initio* data leads to approximate interstitial jump frequencies. Coupling between fluxes is described partially as correlation effects are not accounted for. The recent mean-field developments¹⁵⁵ should be integrated in this type of simulation.

1.08.5.3 Monte Carlo Simulations

AKMC simulations can be used to follow the atomic configuration of a finite-sized system, by performing successive point defect jumps.^{17–19,179,182} In the framework of the chosen interaction model, the migration barriers are exactly computed using equations such as (22) or (23), without any mean-field averaging. The jump to be performed can be chosen with a residence time algorithm,^{183,184} which can also easily integrate creation and annihilation events.^{17,19} Correlation effects between successive point defect jumps, as well as thermal fluctuations, are naturally taken into account. This ensures that diffusion properties and nucleation events are well described (the latter being especially important for the modeling of RIP). The downside is that AKMC simulations are more time consuming than mean-field models, especially when correlation or trapping effects are significant. These advantages and drawbacks explain why AKMC is especially useful to model the first stages of segregation and precipitation kinetics.

AKMC simulations were first used to study RIS and RIP in model systems,^{17,18} to highlight the control of segregation by point defect migration mechanism, and to test the results of classical diffusion equations. These studies show that it is possible – in favorable situations, where correlation and trapping effects are not too strong – to simulate microstructure evolution with realistic dose rates and point defect concentrations, up to doses of typically 1 dpa.

In simple cases, AKMC simulations validate the predictions of continuous models, on the basis of simple diffusion equations^{17–19}: an example is given in Fig. 10 for an ideal solid solution, that is, when RIS can be studied without any clustering or ordering tendency.¹⁷ In this simulation, the diffusion of A atoms by vacancy jumps is more rapid than that of B atoms, and one observes an enrichment of B atoms at the sinks due to the IK effect (A and B atoms diffuse at the same speed by interstitial jumps; those jumps therefore do not contribute to the segregation). One can notice the non-monotonous shape of the concentration

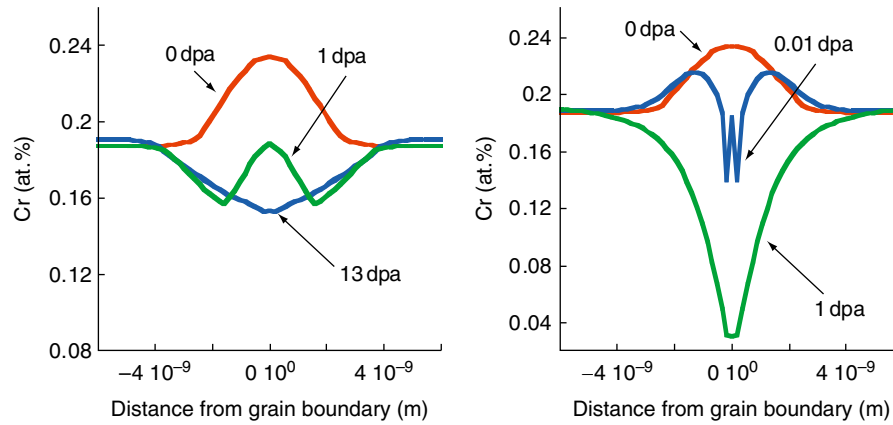


Fig. 9 Comparison of Cr segregation profiles as a function of irradiation dose in FeNi₁₂Cr₁₉ at $T = 635\text{K}$. Left cell represents typical experimental results of Busby *et al.* right cell is mean-field predicted result starting from the experimental profile observed just before irradiation. Reproduced from Nastar, M., 2005. *Philos. Mag.* 85, 641–647. Busby, J.T., Was, G.S., Bruemmer, S.M., Edwards, D.J., Kenik, E.A., 1999. *Mater. Res. Soc. Symp. Proc.* 540, 451.

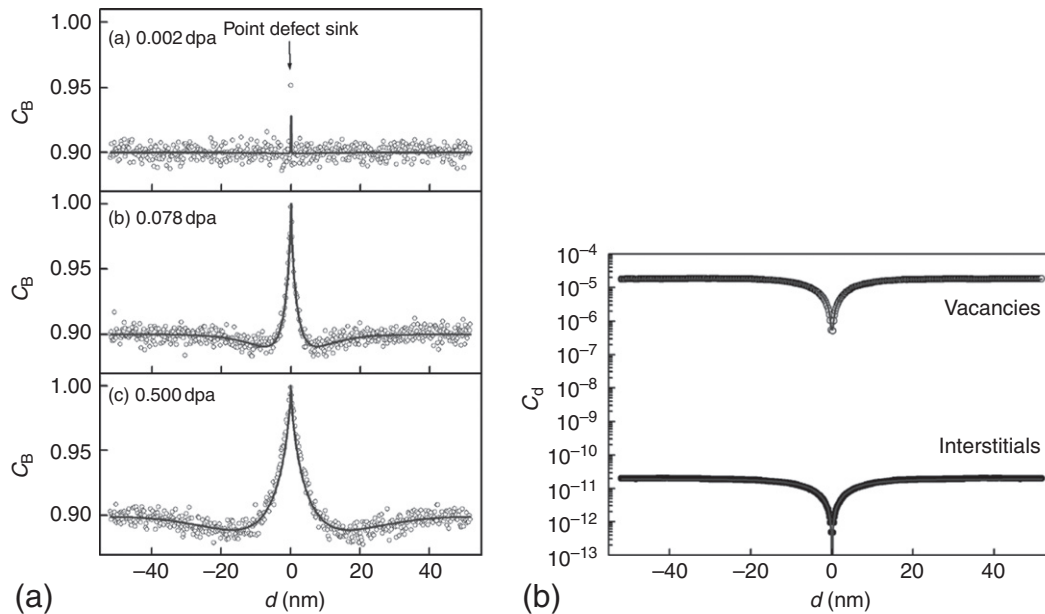


Fig. 10 (a) Evolution of the B concentration profile in an A₁₀B₉₀ ideal solid solution under irradiation at 500K and $10^{-3} \text{ dpa s}^{-1}$, when $d_{AV} > d_{BV}$ and $d_{AI} = d_{BI}$; (b) Point defect concentration profiles in the steady state. Reproduced from Soisson, F.J., 2006. *Nucl. Mater.* 349, 235–250.

profile, which corresponds to the prediction of Okamoto and Rehn³² when the partial diffusion coefficients are $d_{BV}/d_{AV} < d_{BI}/d_{AI}$. In the more complicated case of a regular solution, Rottler *et al.*¹⁸ have shown that the RIS profiles of AKMC simulations can be reproduced by a continuous model, using a self-consistent formulation that gives the dependence of the partial diffusion coefficients with the local composition.

In alloys with a clustering tendency, AKMC simulations have been used to study microstructure evolution when RIS and precipitation interact, either in under- or supersaturated alloys.^{17,19} The evolution of the precipitate distribution can be quite complicated as the kinetics of nucleation, growth, and coarsening depend on both the local solute concentrations and the point defect concentrations (which control solute diffusion), concentrations that evolve abruptly in the vicinity of the sinks.¹⁷ A case of homogeneous RIP, due to a mechanism similar to the one proposed by Cauvin and Martin⁵⁸ (see Section 1.08.2.4), has been simulated with a simplified interstitial diffusion model.^{185,186}

Applications of AKMC to real alloys have been recently proposed. Copper segregation and precipitation in α -iron has been especially studied, because of its role in the hardening of nuclear reactor pressure vessel steels.^{161,179,182,187,188} These studies are based on rigid lattice approximations, using parameters fitted to DFT calculations. The point defect formation energies are much smaller in copper-rich coherent clusters than in the iron matrix,¹⁷⁴ and there is a strong attraction between vacancies and copper

atoms in iron, up to the second-nearest neighbor positions.^{115,174} AKMC simulations show that in dilute Fe–Cu alloys, $L_{CuV} = -(L_{CuCu} + L_{CuFe})$ is positive at low temperatures, because of the diffusion of Cu–V pairs. At higher temperatures, Cu–V pairs dissociate and L_{CuV} becomes negative.^{161,162} The resulting segregation behaviors have been simulated, with homogeneous formation of Frenkel pairs (i.e., conditions corresponding to electron irradiations). Only vacancy fluxes are found to contribute to RIS, the coupling with self-interstitials is negligible. Copper concentration increases near the sinks at low temperatures and decreases at high temperatures.¹⁶¹

Co-precipitation of copper clusters with other solutes (Mn, Ni, and Si) has been modeled by Vincent *et al.*^{179,182} and Wirth and Odette¹⁸⁸ under irradiation at very high radiation fluxes and with formation of point defects in displacement cascades. AKMC simulations display the formation of vacancy clusters surrounded by copper atoms, which could result both from the Cu–V attraction (a purely thermodynamic factor) and from the dragging of Cu by vacancies (effect of kinetic coupling). The formation of Mn-rich clusters is favored by the positive coupling between fluxes of self-interstitials and Mn (DFT calculations show that the formation of mixed Fe–Mn dumbbells is energetically favored).

Similar AKMC simulations have been recently applied to bcc Fe–Cr alloys, as a model of ferritic steels.^{14,20} They provide an estimation of the Onsager coefficients, with their dependence on temperature and composition (Fig. 11(a)). According to Eq. (20), one expects an enrichment of Cr at low temperature ($L_{CrI}/L_{FeI} > L_{CrV}/L_{FeV}$) and a depletion at high temperature ($L_{CrI}/L_{FeI} < L_{CrV}/L_{FeV}$). Direct simulations of RIS at grain boundaries (Fig. 11(b) and (c)) confirm this prediction. These results are in qualitative agreement with those predicted by simpler approximations.^{114,173} It is worth noticing the contributions of self-interstitials and vacancies are quite close, especially above 600K (Fig. 11(a)). The balance between the two competing mechanisms is therefore delicate and small changes in the migration barriers could reverse it. The variability of RIS behaviors observed in ferritic steels probably results from this situation. These results are in qualitative agreement with the continuous RIS models and the experiments of Was *et al.*¹⁷³

The same simulations have been used to study the interaction between precipitate microstructure and RIS in under – as well as in super – saturated alloys.^{14,20} Finally they have shown how RIS decreases with the density of grain boundary.²⁰

1.08.5.4 Phase Field Models

AKMC methods may be very time consuming and are limited to small systems. Phase Field (PF) models are able to tackle larger systems and easily include long-range elastic interactions in addition to the chemical interactions. The PF method is a coarse-grained method relying on a mesoscale diffusion equation. The solid is divided into cells, whose evolution is governed by a mesoscale diffusion equation. When applied to the modeling of flux couplings and RIS, parameterization of the diffusion equation requires a careful methodology of the scale transfer from the PDs jump frequencies to the mesoscale thermodynamic and

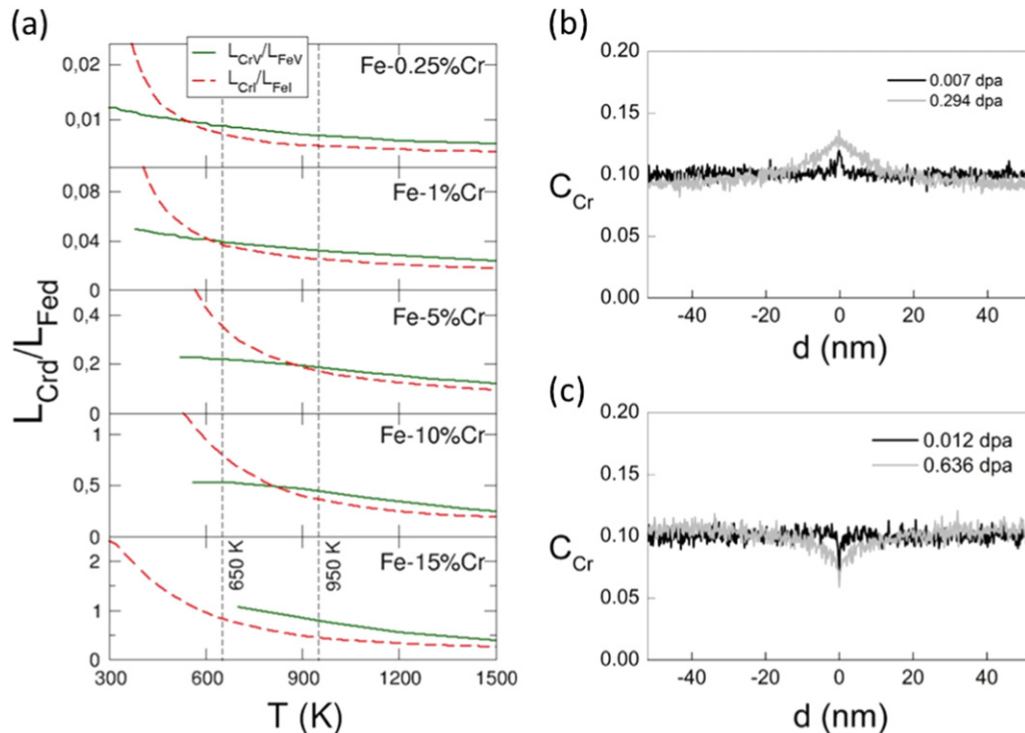


Fig. 11 AKMC simulations of radiation-induced segregation in Fe–Cr alloys Evolution of the fractions of Onsager coefficients L_{CrV}/L_{FeV} and L_{CrI}/L_{FeI} as a function of the temperature and alloy composition (a) and evolution of the Cr concentration profile around a grain boundary in a Fe–10%Cr alloy at (b) 650K and (c) 950K. Reproduced from Senninger, O., Soisson, F., Martínez, E., *et al.*, 2016. *Acta Mater.* 103, 1.

kinetic parameters. This method easily includes mesoscopic features of the microstructure such as point defect sinks including their elastic interaction with point defects. The sink strength of the investigated microstructure features and its effect on the rate of elimination of point defects emerge naturally from the method. We may also introduce equilibrium point defect concentration as a boundary limit of a point defect sink. That way, the latter acts as a source of point defects.^{107,189}

We derive a PF diffusion equation from a TPI-based equation of atomic fluxes similarly to Eq. (4), provided local gradients of chemical potential are derived from a non-uniform free energy functional. This free energy depends on order parameters, which are here local concentrations of chemical species and point defects, and strain fields. The free energy includes a local and a non-local energy term including the long-range elastic interactions between cells. The non-local term (also called gradient energy term) depends on the thermodynamic interface-stiffness parameter and the composition gradients. Note that recently, one has shown kinetic contributions depending on the PD diffusion mechanism may contribute to the non-local term.¹⁹⁰ In principle, the free energy of a Phase Field method must depend on the size of the cell.¹⁷⁵ Accounting for the cell size is crucial whenever one adds to the diffusion equation a stochastic term for the modeling of concentration fluctuations of atomic species or point defect clusters. There are PF methods introducing a mixed discrete-continuous approach to capture the stochastic character of defect cluster production in displacement cascades.¹⁹¹

In case of a monotonic RIS profile spreading over several nanometers, the local energy contribution to the chemical potential mostly contributes to the free energy functional. In general, PF methods applied to RIS ignore the non-local contribution to the free energy.^{107,189} Without the cell size correction and ignoring non-local contributions, the mesoscopic diffusion equation is equivalent to the standard RIS diffusion equation given by Eq. (16). In a standard Cahn–Hilliard diffusion equation, there is a single driving force, the alloy chemical potential gradient, which is the derivative of the alloy free energy with respect to the alloy composition. One implicitly assumes PDs are instantaneously at equilibrium with respect to the local chemical species, which is equivalent to removing gradients of point defect chemical potentials. In systems under irradiation, point defects are not at local equilibrium. Therefore, in addition to the alloy driving force, we include the point defect driving forces and the corresponding off diagonal L -coefficients $L_{V\beta}^V$ and $L_{I\beta}^I$.¹⁰ In dilute systems, one may employ semi-analytical methods to deduce the L -coefficients from *ab initio* data,^{8,139} whereas in concentrated systems one may rely on a computation of the L -coefficients from AKMC simulations.¹⁰⁷ A PF method dedicated to RIS includes a sink and its interaction with all point defects and atoms. Unless the sink is a free surface,¹⁸⁹ one defines a capture shape function of the sink, representing the capture zone of point defects (equal to 0 inside the matrix and 1 in the capture zone). Absorption of point defect may make them move and even change shape. Most studies assume sinks are immobile and do not change, meaning the shape function does not vary under irradiation. Although there are recent PF models simulating the displacement of a free surface,¹⁸⁹ and dislocation climb resulting from a vacancy flow.¹⁹²

One has recently applied the PF method to the specific study of RIS in Fe–Cr. The method relies on a diffusion equation fully parameterized from AKMC simulations including the calculation of the equilibrium point defect concentrations, the thermodynamic factors, the stiffness parameters, and the L -coefficients.¹⁷⁵ The resulting simulations of RIS are in good agreement with direct AKMC simulations. Therefore, we can rely on it to simulate RIS in Fe–Cr of larger size and physical time as compared to those of the AKMC simulations, and introduce the elastic interactions between point defects and sinks.

Point defects may interact with various stress fields from internal sinks or applied stress. One may address these elastic interactions by relying on Eq. (4) and extending the set of order parameters in order to include the shape and elastic characteristics of a sink.¹⁹³ In addition to cell concentrations of all point defects and chemical species, the set of order parameters includes an elastic shape function of every sink associated with an eigen-strain deduced from the micro-elasticity theory. Main difficulty is the computation of the resulting strain which one usually performs in the Fourier space through an iterative procedure. This approach has been applied to the modeling of dislocation sink strengths¹⁹⁴ and RIS resulting from a bias sink strengths between interstitial and vacancy at a dislocation loop in Fe–Cr.¹⁷⁵ Since split-interstitials are more sensitive than vacancy to a stress field, the PD-sink interaction differs between both defects, leading to a stronger interstitial-sink interaction and a larger sink strength associated with interstitial absorption. However, close to the sink the stress field is not uniform and one predicts a biased absorption of vacancy in compressive regions, leading to local changes of sign of RIS in Fe–Cr.¹⁷⁵ Further, one may account for the elastic effect on the L -coefficients relying on the elasto-diffusion formalism,¹³³ and the first principles calculation of the elastic dipoles characterizing the elastic properties of a point defect in a specific local environment.¹⁷⁶ An elasto-diffusion study of vacancy diffusion in dilute Ni(Si) alloys has highlighted the potential change of sign of the flux coupling between vacancy and Si with respect to the strain field and the direction of diffusion. The resulting Si flow near a dislocation core is a complex spatial-and-temperature-dependent function of the vacancy flow. It is mostly due to the interplay between the symmetry reduction of the migrating atom at the saddle point and the elastic field.¹⁰⁹ Incorporating elasto-diffusion into the split-interstitial diffusion mechanism should lead to even more complex flux couplings due to the symmetry break generated by the elastic field between dumbbells with different orientations.

1.08.6 Conclusion

We started this review with a summary of the experimental activity on RIS. Intensive experimental work has been devoted to austenitic steels and their model fcc alloys (Ni–Si, Ni–Cr, and Ni–Fe–Cr) and, more recently, to ferritic steels. RIS strongly depends on the radiation flux and dose, temperature, composition, the grain boundary type, and the point defect sink microstructure. As RIS is a rather deleterious phenomenon, many research teams have focused on reducing it. Addition of large-sized impurities, such

as Zr and Hf, was shown to preclude RIS. Impurities would trap point defects and enhance this way direct recombination reactions between point defects.⁵⁰ However, experimental observations and a systematic KineCluE calculation of flux coupling from DFT-based vacancy jump frequencies tend to show the inhibitory role of most impurities is fading away after a radiation dose of a few dpa. On the other hand, the Cr depletion at grain boundaries was observed to be delayed when the grain boundary was enriched in Cr before irradiation. A 'W-shaped' transitory profile could be maintained until a few dpa before the grain boundary was depleted in Cr.⁴⁰ Even though the development of RIS models was tightly related to the experimental investigations, the underlying mechanisms of an oscillatory profile are still not well understood.

Meantime, the main mechanisms controlling RIS were brought forward before the phenomenon was even observed. From the very first models, one has introduced enhancement of diffusion, flux couplings and point defect driving forces. A RIS model consists of a set of diffusion equations in which we write fluxes of point defects and atoms as generalized Fick's laws. The partial diffusion coefficients of these Fick's laws are function of the phenomenological L -coefficients of TIP.

The formidable progress of the *ab initio* studies addressing point defect properties in alloys and the most recent development of automated calculation codes of the L -coefficients (KineCluE⁸ and Onsager¹³⁹), totally change the situation. First-principles methods relying on the DFT have improved so fast in the last decades that they are able to provide us with large databases of activation energies of both vacancy and interstitial jump frequencies as a function of local environment.^{8,12} Meanwhile, due to recent development of diffusion theories,⁸ it is now possible to overcome the artificial dichotomy between dilute and concentrated diffusion models and for instance, develop a RIS model including a change of sign of the flux coupling with local composition. As a consequence, nowadays, it seems easier to calculate the L -coefficients and their associated partial diffusion coefficients from first-principles calculations rather than estimating them from diffusion experiments. Besides, the recent *ab initio* investigations of RIS in dilute alloys have demonstrated that an extrapolation of diffusion properties from high temperature diffusion experiments almost systematically leads to the wrong sign of the vacancy-solute flux coupling in Fe-based alloys.

While in dilute alloys, diffusion theories yield now efficient codes to address the calculation of the L -coefficients from atomic jump frequencies, in concentrated alloys one employs numerical approaches relying on AKMC simulations or mean field approximate methods. In particular, diffusion models addressing RIS in austenitic steels have lost over the years the proper definition of the partial diffusion coefficients of the Fick's laws in terms of the phenomenological L -coefficients. We explain it by the fact that both experimental diffusion data and *ab initio* investigations of thermodynamic and kinetic point defect properties are still lacking, especially for interstitials. For years, the available diffusion data consisted mainly of tracer diffusion coefficients, and the RIS models still rely on the empirical Manning relations¹⁹⁵ to get an estimation of the partial diffusion coefficients in function of tracer diffusion coefficients.⁶¹

An alternative approach to continuous diffusion equations is the development of atomistic-scale simulations, such as Monte Carlo simulations, which are quite appropriate to study the nanoscale RIS phenomenon. Concerning the ferritic steels, atomistic modeling of RIS in model alloys of ferritic steels has at least contributed to explain the puzzling variety of Cr enrichments and depletions observed experimentally.¹⁴ As already noticed by Choudhury *et al.*,¹¹⁴ RIS of Cr depends on a subtle balance between an enrichment driven by the elimination of self-interstitials and a depletion driven by the elimination of vacancies. Moreover, the Onsager coefficients controlling RIS are very sensitive to small differences in the migration barriers.¹¹¹ Therefore, even though the various models suggest a similar temperature dependence of RIS, the key point is probably that any other factor modifying these barriers (i.e., the magnetic transition, additional alloying elements, elastic stress between sinks and solutes, etc.) may change the final result.

Note that the situation is different in austenitic steels. We do systematically observe positive segregation of Ni and negative segregation Cr whatever the temperature and the composition of alloys. A possible explanation is that self-interstitials and vacancies do not compete; though, the interstitial coupling contribution is still debated. It is rather believed to be either negligible¹⁰⁴ or to reinforce the vacancy contribution,⁶⁰ although a recent study tends to show the opposite.⁶¹

Monte Carlo simulations are now able to embrace the full complexity of RIS phenomena, including vacancy and split interstitial diffusion mechanisms, the whole flux coupling, the resulting segregation, and eventual nucleation at grain boundaries.¹⁴ However, these simulations become heavy time-consuming when correlation effects are important. Using this method or atomic diffusion theories to feed a PF method may be the most promising approach. One has successfully achieved it to simulate RIS in Fe–Cr.¹⁰⁷ A PF method gives new insights on the isolated contribution to RIS of each physical parameter of the diffusion equation and on the interplay between the elastic point defect – sink interaction and RIS.¹⁹³ In the next future, we should be able to encompass the entire microstructure and its interaction with all point defects and atoms, provided the stochastic nature of point defect cluster and secondary phase precipitation are included in a proper way.

Acknowledgments

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See also: 1.04 Radiation-Induced Effects on Microstructure. 1.05 Effect of Radiation on Strength and Ductility of Metals and Alloys. 1.13 Radiation Damage Correlation. 1.25 Phase Field Methods. 1.26 Computational Thermodynamics: Application to Nuclear Materials. 1.27 Computational Kinetics: Application to Nuclear Materials

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